NANYANG TECHNOLOGICAL UNIVERSITY

MULTIMODAL SPEECH EMOTION RECOGNITION WITH LARGE LANGUAGE MODELS

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College of Computing and Data Science

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Abstract

Speech Emotion Recognition (SER) is a critical component of human-computer interaction, with significant applications in fields such as mental health monitoring, customer service, and security surveillance. However, traditional SER approaches face two primary limitations: they often evaluate utterances in isolation and rely on coarse taxonomies and fail to capture fine-grained emotions. While the emergence of Large Language Models (LLMs) has demonstrated superior capabilities in semantic reasoning, applying them for SER task are not without limitations. Specifically, prominent Large Audio Language Models (LALMs) such as SALMONN and Qwen2-Audio rely on internal, continuous speech encoders. This inherently opaque, "black-box" mapping process lacks the interpretability, which is crucial for specialized domains. To address this limitation, this report builds upon the SpeechCueLLM baseline, which discretizes numeric speech features into natural language descriptors to leverage the LLM's inherent reasoning abilities.

We propose an enhanced framework that integrates a structured prompting strategy, a specialized acoustic feature set, and Dynamic Feature Masking (DFM) to improve both classification performance and environmental robustness. We utilize the extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPS) to capture granular paralinguistic cues, including spectral and voice quality features. Furthermore, the continuous affective dimensions of Valence, Arousal, and Dominance (VAD) are integrated to provide high-level emotional context. To manage the high dimensionality of these acoustic parameters, we employ a hybrid feature selection strategy combining Lasso Regression and ANOVA, ultimately mapping the most discriminative features into discrete, quantile-based natural language bins.

The proposed framework is rigorously evaluated across two distinct acoustic domains: the acted, dyadic interactions of IEMOCAP and the naturalistic, "in-the-wild" speech of MSP-Podcast. On the IEMOCAP corpus, the model achieves a Macro-F1 score of 72.0%, representing a 1.3% absolute improvement over the foundational SpeechCueLLM baseline. More notably, on the highly challenging, 16-class MSP-Podcast dataset, the framework achieves a substantial 6.5% absolute increase, reaching a Macro-F1 score of 25.1%. Across both domains, the proposed architecture establishes new state-of-the-art results, consistently outperforming both self-supervised audio representation models and zero-shot multimodal LLM competitors.

Ultimately, these results demonstrate the pipeline's versatility in effectively processing both prototypical primary emotions and complex, spontaneous affective states without relying on human-annotated metadata. By discretizing acoustic cues into human-readable text, the framework not only achieves superior predictive performance but also establishes a highly interpretable and justifiable reasoning pathway over black-box models.

Acknowledgments

I would like to express my deepest gratitude to my academic advisor, Prof. Chng Eng Siong, for providing me with the opportunity to undertake this research and for his/her continued support throughout this journey.

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Chapter 1

Introduction

Human communication relies heavily on the paralinguistic nuances of speech to convey complex emotional states, making it inherently multimodal. Speech Emotion Recognition (SER) aims to computationally identify these affective dimensions to facilitate more intuitive Human-Computer Interaction (HCI). However, current popular systems present various challenges in contextual reasoning, and the adoption of fine-grained affective taxonomies. This chapter establishes the foundation of the thesis by outlining the motivation for advancing SER via Large Language Models (LLMs). It proceeds to define the specific research objectives, project scope, and the novel contributions of the proposed automated framework. Finally, Section 1.6 provides an overview of the report's structure.

1.1 Background

Emotion Recognition (ER) is a pivotal component of HCI, enabling systems to perceive and respond to user affective states. This technology has broad downstream applications, particularly in mental health monitoring [1], automated customer service [2], and security surveillance [3]. For instance, Gaikwad and Venkatesan [4] proposed a methodology to assess depression by analyzing emotional and paralinguistic cues embedded in spoken language. Similarly, in the customer service domain, ER systems are employed to evaluate satisfaction by dynamically recognizing user sentiment [2], while

in healthcare, such systems support the non-invasive detection of neurodegenerative conditions like Alzheimer's disease [5].

Human emotion is inherently multimodal, conveyed through diverse channels including speech, textual content, and facial expressions [6]. While physiological signals such as Electroencephalogram (EEG) brain waves serve as an alternative source for affective analysis [7], the primary non-invasive modalities remain acoustic and linguistic. In the acoustic domain, prosodic, spectral, and voice quality features—such as pitch, formants, MFCCs, and jitter—are widely utilized to capture paralinguistic information [8]. Complementing this, linguistic features derived from textual transcripts provide necessary semantic context regarding the speaker's intent and the situational causality of the emotion.

Speech Emotion Recognition (SER) is formally defined as the computational task of identifying and categorizing human emotional states directly from these speech signals. Recently, driven by the rapid emergence and advancement of Large Language Models (LLMs), there has been a paradigm shift towards adapting these generative models for affective computing tasks. Consequently, this research focuses on **Multimodal SER**, integrating both acoustic and linguistic modalities within an LLM-based framework to achieve robust and context-aware inference.

1.2 Challenges and Limitations

ER remains a challenging endeavor due to the inherent ambiguity and complexity of human affect. Ahmed et al. [9] observe that unimodal models often struggle to capture the comprehensive emotional state of a user, thereby necessitating the adoption of multimodal approaches. However, the non-linear interaction between these modalities introduces significant computational complexity [9]. Furthermore, while Deep Neural Networks (DNNs) excel at modeling these complex multimodal interactions [10], their "black-box" nature results in a lack of explainability. This opacity severely hinders their applicability in sensitive domains such as healthcare, where interpretable decision-making is critical [11].

1.2.1 Limitations of Text Emotion Recognition (TER)

While textual content provides direct semantic information, relying solely on linguistic transcripts is often insufficient for accurate emotion detection. TER models frequently struggle with tonal ambiguity, failing to detect sarcasm, irony, or subtext where the spoken intonation contradicts the literal semantic meaning [12].

1.2.2 Limitations of SER

Current approaches in SER face three primary limitations concerning contextual awareness, emotional granularity, and model interpretability:

Lack of Contextual Awareness: Many State-of-the-Art (SOTA) models, such as emotion2vec [13], perform evaluation at the level of a single utterance. This approach overlooks the temporal dynamics of conversation, failing to account for how emotions transition or intensify over the course of a dialogue [14].

Limited Emotion Taxonomy: A significant limitation of publicly available models and datasets is their exclusive focus on coarse-grained, primary emotions (e.g., Happy, Sad, Angry) [13, 15]. This restricts their effectiveness in naturalistic, real-world scenarios—such as podcasts or therapeutic sessions—where subtle secondary states like "Concerned" or "Disappointed" are often more prevalent than exaggerated primary emotions [16].

1.3 Objective

The primary aim of this project is to develop a robust, fully automated Multimodal SER framework capable of processing raw audio inputs for real-world applications. Unlike traditional systems that rely on manual transcription or preprocessed feature sets, the proposed model functions as an automated pipeline—converting raw audio into structured prompts for LLM inference without human intervention. To satisfy the demands of real-world applications, the architecture accommodates highly scalable affective taxonomies through dataset-specific optimizations.

The specific research objectives are as follows:

- **To Develop a Context-Aware Dialogue System:** To enable the analysis of long-form audio conversations by integrating historical context into the decision-making process. This ensures the model can resolve ambiguity in individual utterances by conditioning predictions on the preceding dialogue flow.
- **To Design a Universal Framework:** To formulate a flexible prompting strategy that allows the SER system to adapt to various datasets and emotion taxonomies (primary or secondary) without requiring structural modifications to the underlying LLM architecture.
- **To Achieve Cross-Corpus Generalization:** To validate system robustness by alternately training and testing the framework on studio-acted data (IEMOCAP [15]) and naturalistic data (MSP-Podcast [16]), to ensure that the learned features are highly transferable rather than dataset-specific.
- **To Benchmark against SOTA:** To quantitatively evaluate the proposed framework against existing off-the-shelf baselines—including both unimodal (audio-only) and multimodal approaches—to demonstrate the efficacy of LLM-driven reasoning in ER.

1.4 Project Scope

This research frames SER as a closed-set classification task leveraging the generative capabilities of LLMs. The core methodology utilizes an LLM to infer and generate specific emotion labels (e.g., "Angry", "Sad") based on structured text prompts derived from extracted acoustic and linguistic features.

The study is subject to the following boundaries:

- **Modality Focus & Prioritization:** While the system incorporates textual transcripts, the research prioritizes Multimodal SER. The acoustic modality (prosody, spectral, voice quality) is treated as a critical decision-making component to resolve semantic ambiguity.

- **Language Constraints:** The scope is strictly limited to the English language. All pre-trained models and datasets utilized are English-based; cross-lingual emotion transfer is not covered.
- **Data Domain:** The experimental evaluation focuses on two distinct acoustic domains:
  1. **Studio/Acted Speech:** Represented by the IEMOCAP dataset [15], characterized by high-fidelity audio and dyadic interactions.
  2. **Naturalistic/Spontaneous Speech:** Represented by the MSP-Podcast dataset [16], characterized by realistic background noise and fine-grained emotional taxonomies.

Out-of-Scope

The following areas are explicitly excluded from the current study:

- **Open-Set Emotion Generation:** While LLMs are capable of generating open-ended natural language descriptions of emotions, this project restricts the model to a predefined taxonomy of emotion classes. The model is not evaluated on its ability to generate novel or zero-shot emotion labels outside of this target set.
- **Visual Modality:** Although human emotion is inherently multimodal, this research excludes visual cues such as facial expressions and gestures. The focus is strictly placed on auditory (paralinguistic) and linguistic (textual) modalities.
- **Continuous Dimensional Regression:** A common alternative in SER is predicting continuous values for Valence, Arousal, and Dominance (VAD) [17]. This project does not frame the task as a regression problem. Instead, VAD values are utilized as input features within the prompt to aid the LLM's reasoning process, rather than serving as the target output.

1.5 Contributions

This research advances the field of Multimodal SER by proposing a robust, LLM-driven framework capable of operating in naturalistic acoustic environments. The primary contributions of this work are summarized as follows:

- **Development of an Multimodal Framework:** We propose a fully automated pipeline that eliminates the reliance on manual annotations (transcriptions and speaker metadata) required by prior baselines. By integrating Automatic Speech Recognition (ASR) and gender classification modules, the system enables practical, real-time inference from raw audio inputs.
- **Structured Prompting with Extended Acoustic Representation:** We introduce a novel prompting strategy that replaces natural language descriptors with a structured feature table. Unlike the baseline's limited prosodic set, our framework incorporates the eGeMAPS. We employ a hybrid feature selection mechanism (combining Lasso Regression and ANOVA) to identify up to 30 most discriminative parameters, enhancing the LLM's ability to resolve semantic ambiguity.
- **Dynamic Feature Masking (DFM) Strategy:** To mitigate overfitting and encourage robust cross-modal reasoning, we implement a DFM technique. This regularization strategy incrementally masks acoustic tokens during training, preventing the model to over-rely on any single modality.
- **SOTA Performance and Cross-Domain Robustness:** We provide a comprehensive evaluation on both acted (IEMOCAP) and naturalistic (MSP-Podcast) datasets. The proposed framework establishes strong benchmarks for LLM-based SER, achieving a 1.3% absolute increase in Macro-F1 on IEMOCAP and a substantial 6.5% improvement in Macro-F1 on the highly ambiguous MSP-Podcast, achieving new SOTA results, and outperforms both self-supervised ER models and zero-shot multimodal LLM competitors. Furthermore, rigorous zero-shot cross-corpus evaluations validate the framework's capability to generalize learned paralinguistic representations across disparate acoustic domains.

- **Interpretable and Justifiable Reasoning Pathway:** Unlike conventional black-box models, the proposed framework establishes a transparent structural advantage by explicitly discretizing numeric speech features into natural language bins. This design inherently ensures that the model's affective predictions are justifiable and grounded in human-readable acoustic phenomena, bypassing the opacity of traditional deep learning encoders.

1.6 Report Structure

This report consists of five chapters:

- **Chapter 1: Introduction** - Provides an overview of the research landscape, identifying the limitations of current SER systems and establishing the motivations, objectives, and scope of the proposed multimodal LLM-based framework.
- **Chapter 2: Literature Review** - Surveys the evolution of multimodal SER, ranging from traditional acoustic feature analysis to SOTA Deep Learning approaches, with a specific focus on recent advancements in LLMs.
- **Chapter 3: Methodology** - Deep dive to the baseline methodology, highlight its limitations, and details the theoretical design of the proposed framework, explaining the feature extraction pipeline, the prompt construction strategy, and the dynamic feature masking strategy.
- **Chapter 4: Experimental Setup** - Describes the implementation details, including the preprocessing of IEMOCAP and MSP-Podcast datasets, the specific hyperparameter configurations for fine-tuning, and the evaluation metrics used. Furthermore, it establishes the benchmarking baselines and outlines the methodological design of the ablation studies used to isolate component contributions.
- **Chapter 5: Result Analysis** - Presents the quantitative and qualitative findings of the study. It offers a critical analysis of the framework's performance across varied datasets and fine-grained emotion classes, followed by comprehensive ablation studies and error propagation analysis.

- **Chapter 6: Conclusion & Future Discussion** - Summarizes the key contributions and findings of the research, discusses the limitations encountered, and proposes potential directions for future improvements in multimodal SER.

Chapter 2

Literature Review

The field of affective computing has evolved rapidly, transitioning from traditional statistical acoustic models to sophisticated deep learning approaches. To contextualize the proposed methodology, this chapter surveys the multimodal Emotion Recognition (ER) model. Section 2.1 and Section 2.2 provide overviews of Text Emotion Recognition (TER) and Speech Emotion Recognition (SER), respectively. Building upon this, Section 2.3 explores strategies for integrating these independent modalities. Finally, Section 2.4 examines the recent paradigm shift toward utilizing Large Language Models (LLMs) for generative ER, detailing strategies for acoustic-to-text modality alignment and reviewing the emergence of instruction-tuned multimodal foundation models.

2.1 TER

TER classifies emotions based solely on textual data, utilizing Natural Language Processing (NLP) models to interpret semantic content. Bidirectional Encoder Representations from Transformers (BERT) has become a cornerstone architecture in this domain due to its superior capacity for contextual understanding [18]. For instance, Kumar and Raman [19] utilized BERT to extract word embeddings, which were subsequently processed through a crossed dual-channel architecture. By processing the embeddings through parallel LSTM-CNN and CNN-LSTM pathways, the framework effectively captures both local features and long-range sequential dependencies (Figure 2.1).

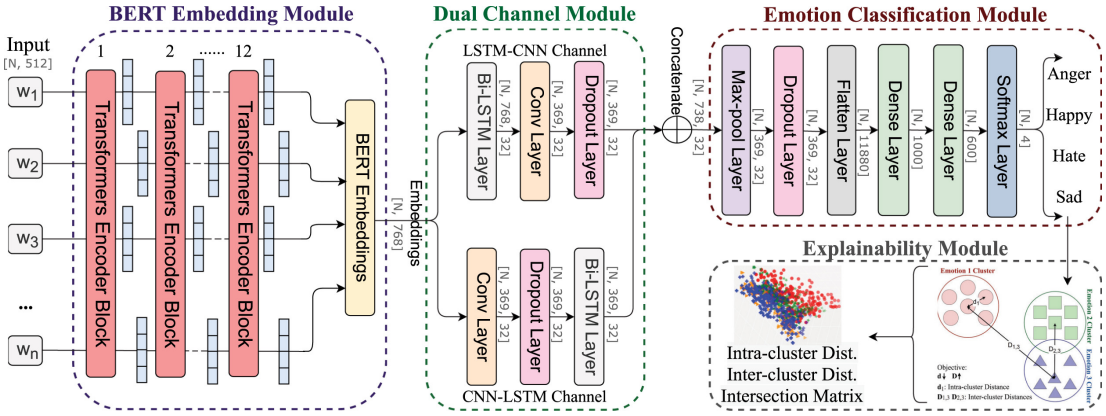

Figure 2.1: Dual Channel TER: Utilizing parallel LSTM-CNN and CNN-LSTM pathways for local and temporal feature extraction.

Beyond the analysis of isolated sentences, TER frameworks are particularly effective at modeling dynamic conversational contexts. For example, Majumder et al. [14] designed an Recurrent Neural Networks (RNN)-based system capable of distinguishing between speakers (speaker modeling) to better track emotional shifts across a dialogue.

2.2 SER

SER is inherently an interdisciplinary domain, converging linguistics, psychology, and computer science. This section traces the evolutionary trajectory of acoustic emotion recognition, beginning with foundational statistical models and transitioning into the deep neural network architectures that define current state-of-the-art (SOTA) systems.

2.2.1 Traditional SER

Early approaches addressed the task using probabilistic Gaussian Mixture Models (GMMs). Subsequently, Hidden Markov Models (HMMs) were employed to better model the temporal dynamics inherent in speech [20, 21]. These traditional paradigms relied heavily on extensive feature engineering and selection, focusing primarily on low-level descriptors such as pitch, energy, and voice quality [22]. Consequently, Support Vector Machines (SVMs) became widely adopted during this era due to their effectiveness in handling high-dimensional, hand-crafted acoustic features [23].

2.2.2 Neural Network-Based SER

The field later shifted toward neural network architectures, which dominate current research. Unlike traditional methods, deep neural networks require minimal feature engineering and rely on large-scale datasets to learn representations automatically. Notable architectures include RNNs [24] and CNNs [25]. The introduction of attention mechanisms further advanced the field; for instance, Xie et al. [26] utilized attention-based LSTM to weight the importance of specific speech frames dynamically. More recently, Self-Supervised Learning (SSL) has emerged as the SOTA paradigm. Models such as Wav2Vec 2.0 (W2V2) [27] are pre-trained on vast unlabeled corpora and subsequently fine-tuned on task-specific datasets like IEMOCAP [15]. Emotion2vec [13] (Figure 2.2) utilize this W2V2-based SSL architecture to yield robust emotion recognition performance.

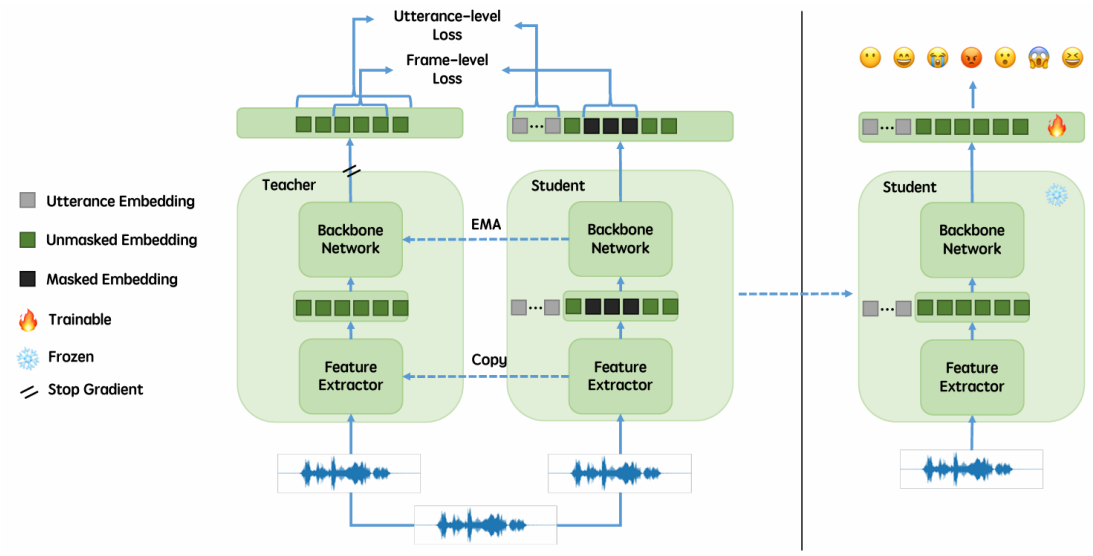

Figure 2.2: Emotion2vec Framework: A self-supervised Teacher-Student pre-training strategy and subsequent downstream fine-tuning for emotion classification.

2.3 Multimodal Emotion Recognition Model

Recent research increasingly combines audio and text modalities to enhance emotion recognition performance. To leverage both modalities simultaneously, their respective feature sets must be effectively integrated. Common fusion strategies include feature-level fusion (early fusion) [28] and decision-level fusion (late fusion) [29]. Early

fusion involves concatenating acoustic and textual features into a single vector prior to processing, whereas late fusion trains separate unimodal models and aggregates their independent predictions via voting or averaging mechanisms [9]. Beyond these traditional methods, more advanced techniques employ attention-based fusion; for example, Tsai et al. [30] utilize a Transformer encoder to integrate signals via cross-modal attention mechanisms, allowing the model to weigh the interaction between modalities dynamically.

2.4 LLMs for Emotion Recognition

Building upon the multimodal fusion strategies discussed previously, the advent of Large Language Models (LLMs) has fundamentally transformed Affective Computing. This section explores the paradigm shift toward generative emotion recognition, detailing how LLMs are utilized for conversational context tracking, examining strategies for integrating continuous speech features into discrete textual architectures, and reviewing the emergence of foundation audio models.

2.4.1 Emotion Recognition in Conversation (ERC)

The rapid advancement of LLMs has revolutionized the sub-domain of ERC. Recent literature demonstrates that LLMs possess an inherent capability to perceive and reason about human emotional states [31]. Unlike traditional statistical models, LLMs process textual input through complex cognitive-like mechanisms, allowing them to interpret semantic nuances akin to human reasoning [32].

This reasoning capability has been leveraged in various capacities. For instance, Dutta and Ganapathy [33] utilized LLMs as automated annotators to generate pseudo-labels for unlabeled ASR transcripts, successfully supervising the training of smaller, resource-constrained TER models. Building on this, recent works have shifted the paradigm from discriminative classification to generative emotion recognition. InstructERC (Figure 2.3) [34] formulates ERC as a generative instruction-following task, demonstrating that LLMs could outperform specialized discriminative models by di-

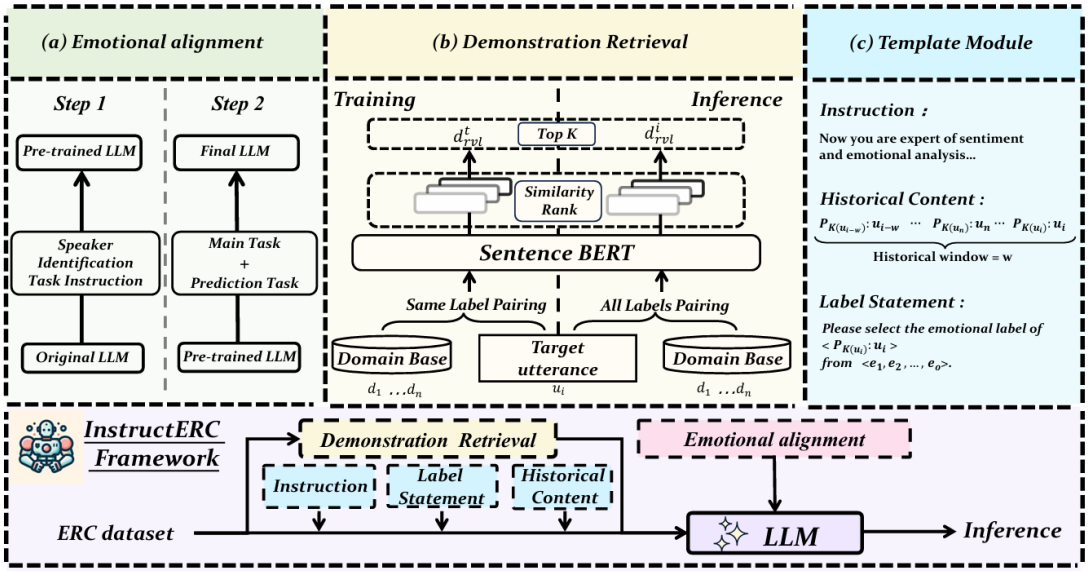

Figure 2.3: InstructERC Framework: Reformulating emotion recognition in conversation as a generative task utilizing retrieved demonstrations and structured instruction templates

rectly generating emotion labels as text. Furthermore, this generative approach has expanded multimodally; DialogueLLM [35] integrated visual modalities by converting video frames into textual descriptions. Feeding these descriptions into the prompt enabled multimodal inference within a purely text-based architecture influences the methodology of this project.

2.4.2 Speech Incorporation in LLMs

Integrating speech features into LLMs remains an active area of research, with methodologies generally falling into prompt-based or architectural adaptation strategies. As the foundational baseline for this work, SpeechCueLLM [36] proposes the discretization of numeric acoustic features into natural language "bins". These descriptive bins are provided as part of the text prompt alongside the audio transcription, enabling the LLM to infer emotion tags without requiring architectural modification.

A comparable approach was proposed by Santoso et al. [37], who similarly incorporated speech feature descriptors as natural language text within the prompt. However, they extended this methodology by utilizing the LLM-generated labels to supervise the training of a smaller, distilled model. Their results demonstrated that these weakly

generated labels could achieve performance comparable to human annotation at a significantly lower cost.

Alternative strategies focus on architectural adaptation rather than prompting. Fig. 2.4 shows the framework of SECap [38], where it employs a HuBERT audio encoder to encode raw audio, and utilizes a bridging network to project these vector representations into a latent space compatible with LLaMA. This allows the model to directly decode and generate speech emotion captions. Similarly, Zhang et al. [39] introduced the use of acoustic landmarks as specialized input tokens, enabling the LLM to detect depression directly from spectral patterns.

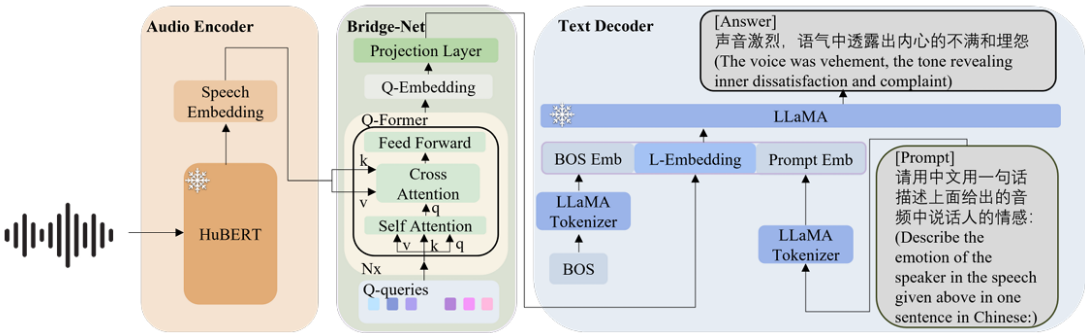

Figure 2.4: SECap Framework: Utilizing a HuBERT audio encoder and a Q-Former bridging network to align continuous audio representations with LLaMA for text generation

2.4.3 Instruction-Tuned Foundation Multimodal LLMs

Beyond specialized SER architectures, a substantial body of research focuses on instruction-tuned foundation models trained across diverse audio tasks, including emotion recognition. A prominent subclass of these are Large Audio Language Models (LALMs), which leverage shared representations across varying auditory domains to yield competitive performance on downstream tasks. Notable LALM examples include SALMONN [40], which specializes in general audio perception and paralinguistic analysis, and Qwen2-Audio [41] (Figure 2.5), which extends these capabilities into the conversational domain. Building upon these LALM foundations, broader multimodal architectures like Qwen3-Omni [42] facilitate end-to-end multi-sensory speech interaction and complex affective reasoning.

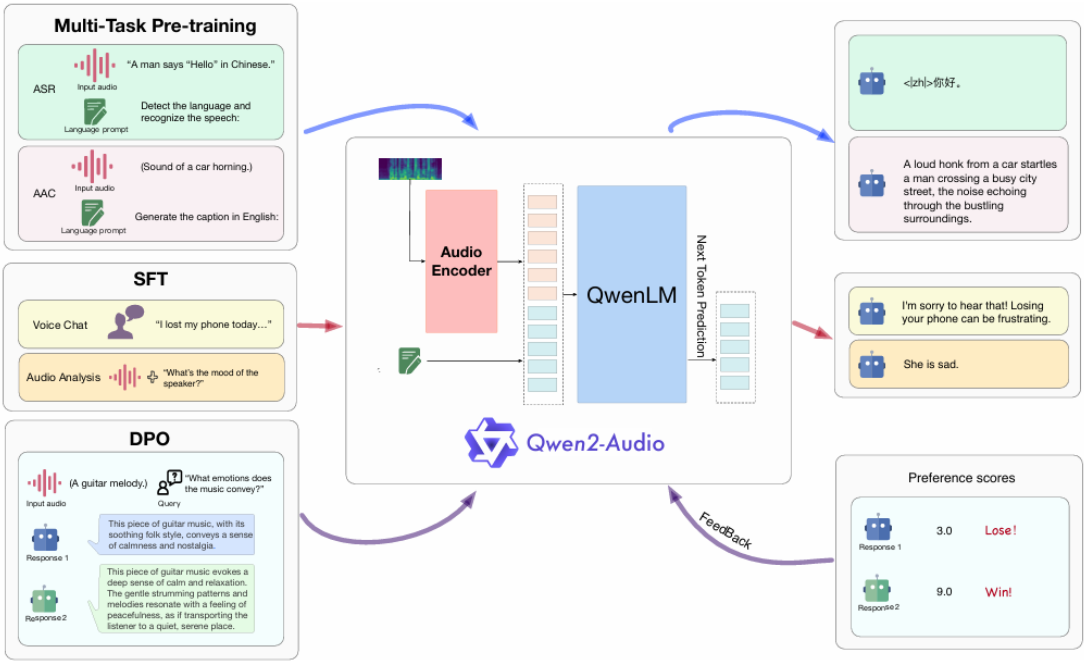

Figure 2.5: Training Pipeline of Qwen2-Audio: Illustrating the progression from multi-task pre-training to Supervised Fine-Tuning for robust audio-linguistic alignment.

Chapter 3

Methodology

This chapter details the methodological design of the research by contrasting the proposed framework against the established SpeechCueLLM baseline. It begins with Section 3.1, which deconstructs the SpeechCueLLM baseline, outlining its feature extraction, discretization, and prompt construction processes. Subsequently, Section 3.2 critically examines the limitations of this baseline, specifically highlighting its reliance on manually annotated data, restricted emotion taxonomies, and a limited set of acoustic features that induces a "semantic bottleneck". To address these challenges, Section 3.3 presents the proposed framework. This section details the integration of the high-dimensional eGeMAPS, the implementation of statistical feature selection and Stochastic Valence, Arousal, Dominance (VAD) Substitution, and the introduction of Dynamic Feature Masking (DFM). Finally, it outlines the transition to a Structured Key-Value Prompting strategy designed to maximize LLM reasoning efficiency across diverse acoustic domains.

3.1 Baseline Preliminary

SpeechCueLLM [36] serves as the foundational baseline for this research. Its core methodology involves converting numerical speech features into discretized natural language descriptions (bins), thereby leveraging the inherent Natural Language Understanding (NLU) capabilities of LLMs to perform emotion recognition solely from tex-

tual input. Crucially, this approach enables the processing of multimodal data through prompt engineering, eliminating the need for complex architectural modifications to the LLM.

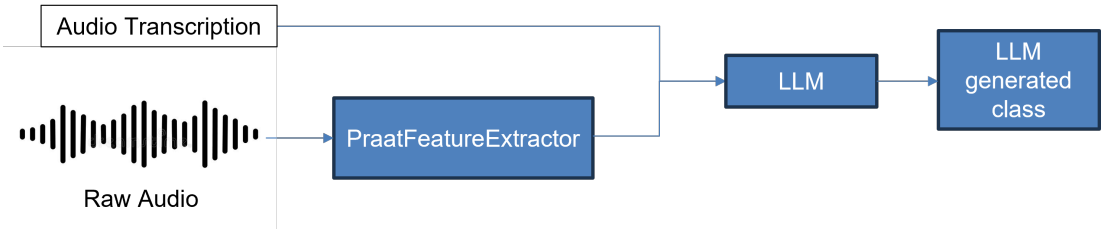

Figure 3.1: SpeechCueLLM Baseline Pipeline

Figure 3.1 illustrates the pipeline. The system utilizes Praat [43] to extract acoustic features, which are subsequently integrated into the LLM prompt alongside the corresponding audio transcriptions. The model then generates the a single emotional label from its predefined taxonomy as a text output.

3.1.1 Feature Extraction and Discretization

Praat is a widely used, open-source software tool to extract speech features such as pitch and intensity. The baseline system utilise praat toolkit to extract five core acoustic features to characterize the prosodic nature of the audio:

- **Average Volume:** Represents the overall loudness or intensity of the speech.
- **Volume Variation:** Captures the dynamic range and fluctuations in intensity (standard deviation).
- **Average Pitch:** Indicates the general fundamental frequency (F_0) of the speaker's voice.
- **Pitch Variation:** Measures the modulation or variability in pitch (standard deviation).
- **Speaking Rate:** Quantifies the speed of speech (syllables per second excluding pauses).

To bridge the modality gap between continuous numerical data and the text-based LLM, the system employs quantile-based discretization. Specifically, continuous values are

mapped into five distinct quintiles, corresponding to the natural language categories: *Very Low, Low, Medium, High,* and *Very High*.

Captions:
No, they told me-I'm sorry, but they told me to be in this line.

Perceptual Analysis:
The speaker conveys a distinct sense of restrained frustration and defensive assertion. The delivery begins with a sharp, impulsive spike on the initial denial (No, they told me...")—suggesting immediate annoyance—before quickly settling into a more controlled, socially acceptable tone during the apology (I'm sorry..."). The voice sounds firm and insistent rather than submissive, yet it stops short of shouting or overt aggression.

Description:
Target speech characteristics: moderate pitch with high variation, high volume with moderate variation, very low speaking rate.

Figure 3.2: SpeechCueLLM Description Example

Based on these categorical values, the system generates descriptive natural language phrases tailored for LLM consumption. An example of such a speech description is illustrated in Figure 3.2.

3.1.2 Prompt Construction

To fully utilize the In-Context Learning (ICL) capabilities of LLMs, the baseline incorporates a sliding window mechanism with a context width of N. This captures the preceding N conversational utterances as historical context, which are processed alongside the target utterance (total window size of $N + 1$) to inform the decision-making process. Additionally, pitch features are explicitly appended to the end of the three most recent utterances within the prompt. Empirical analysis by the baseline authors suggests that pitch dynamics in the immediate conversational history yield the highest correlation with accurate emotion prediction [36].

An example of the constructed prompt is presented in Figure 3.3.

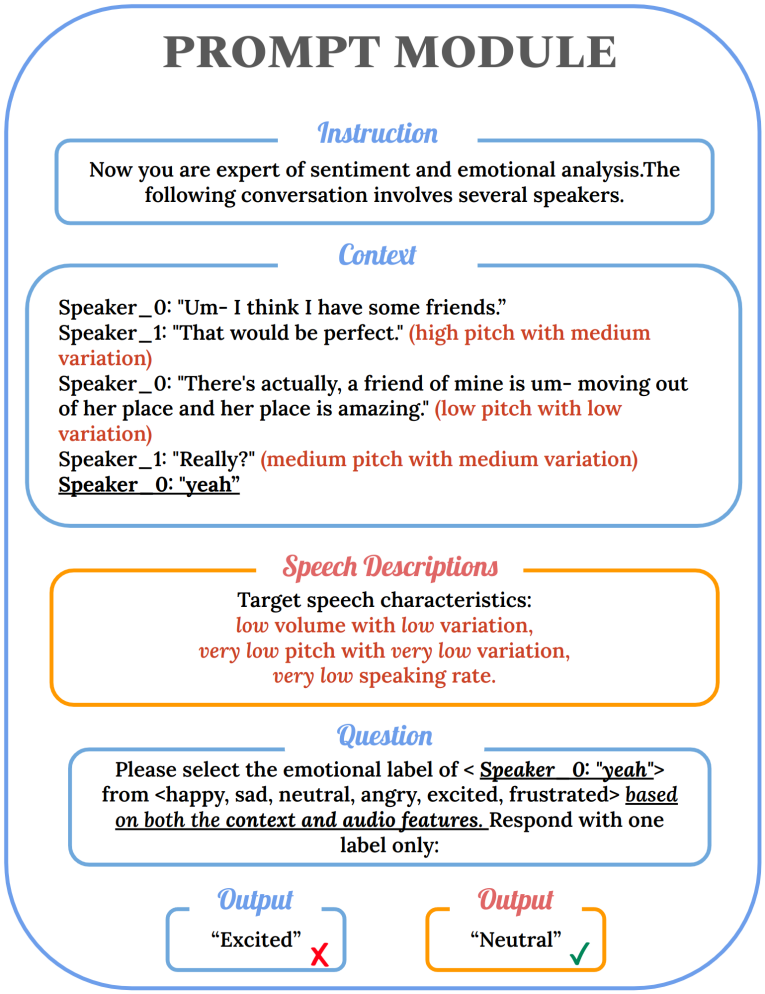

Figure 3.3: Baseline Prompt

3.2 Limitations of the Baseline

While SpeechCueLLM establishes a strong foundation for LLM-based SER, several limitations hinder its direct applicability to real-world deployment: **Reliance on manual Annotations:** A critical limitation is the baseline's reliance on pre-provided ground-truth data, specifically manual transcriptions and gender labels, alongside the raw audio. This dependence highlights a gap between controlled research conditions and practical deployment, where such metadata is rarely available in real-time.

Restricted Emotion Taxonomy: The baseline is constrained to recognizing only six emotions. This taxonomy fails to encompass the full spectrum of primary emotions, nor does it account for fine-grained secondary states (e.g., "Disappointed", "Concerned"),

as humans are capable of understanding and recognising more than six emotions [44].

Unexplored Single-Utterance Performance: While the utilization of historical context via ICL is a key strength, the baseline's efficacy in single-utterance scenarios remains underexplored. Evaluating performance without conversational history is crucial for applications where long-form context is unavailable.

Feature Selection and Prompt Efficiency: The baseline's reliance on a limited set of five acoustic features captures only macro-level dynamics, failing to distinguish emotions with similar prosodic envelopes. For instance, "Angry" and "Excited" both exhibit high pitch and intensity, creating a "semantic bottleneck" where they become overlapping in feature space without finer spectral analysis. While linguistic context offers some disambiguation, it remains insufficient due to the inherent ambiguity of natural text. However, naively expanding the feature set risks "prompt bloat", where excessive token consumption dilutes the LLM's attention mechanism and degrades reasoning efficacy.

Prompt Structure Optimization: While the baseline authors explored adding more descriptive features, they overlooked a fundamental aspect of prompt engineering: the organization of information. The comparison between unstructured natural language descriptions and structured representations remains an open research question.

3.3 Proposed Methodology

The baseline framework presents a computationally efficient foundation by avoiding complex architectural modifications to the LLM. Its method of incorporating speech features via natural language is highly scalable and offers inherent interpretability, as the model conditions its predictions on human-readable descriptions. Building upon these strengths, this section details the proposed enhancements designed to address the identified limitations.

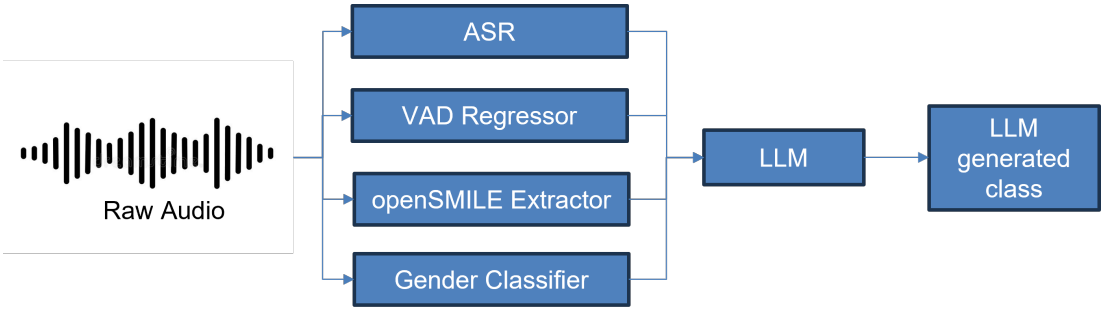

Figure 3.4: SpeechCueLLM Full Pipeline

3.3.1 System Overview

Figure 3.4 illustrates the revised architecture. Unlike the baseline, this framework operates as a fully automated pipeline requiring only raw audio input, thereby eliminating the dependency on manual transcriptions and metadata during inference.

The system comprises four primary components:

1. **ASR Module:** Automatically generates transcriptions from raw audio in real-time.
2. **VAD Regression Module:** Predicts continuous dimensional scores for Valence, Arousal, and Dominance (VAD) directly from the audio.
3. **Extended Feature (OpenSMILE) Extractor:** Extracts a comprehensive acoustic feature set beyond standard prosody, incorporating spectral descriptors (e.g., MFCCs) and voice quality metrics (e.g., Jitter). These signals capture vocal timbre and physiological tension, resolving ambiguities in high-arousal states (e.g., "Angry" vs. "Excited") where pitch and volume are insufficient.
4. **Gender Classifier:** A classifier trained on raw audio to automatically infer speaker gender.
5. **LLM Classifier:** The central inference engine that integrates these multimodal signals to generate the final emotion tag.

Operationally, the ASR and acoustic processing modules run in parallel. Their outputs–transcriptions, VAD scores, gender labels, and extended acoustic descriptors–are subsequently fused into a structured prompt and fed to the LLM for final classification.

3.3.2 Speech Features Incorporation

eGeMAPS

Basic prosodic descriptors such as pitch, loudness, and speaking rate are insufficient to encompass the full spectral and temporal dimensionality of complex emotional speech. To address this limitation, the proposed framework adopts the eGeMAPS (Geneva Minimalistic Acoustic Parameter Set) [45], extracted using openSMILE toolkit [46]. Unlike the baseline, which relies on a limited set of heuristic descriptors, eGeMAPS extracts a standardized vector of 88 acoustic parameters explicitly optimized for paralinguistic analysis.

A critical distinction between the baseline extraction methodology (Praat toolkit) and our Proposed Extended Feature Set (eGeMAPS) lies in the depth and perceptual relevance of the acoustic analysis:

- **Perceptual Pitch Scaling:** While Praat toolkit typically computes Fundamental Frequency (F_0) on a linear Hertz scale, eGeMAPS transforms pitch into a logarithmic semitone scale. This aligns more closely with human auditory perception, where relative melodic contours are more significant for emotion perception than absolute frequency values.

- **Spectral Envelope & Vocal Tract Dynamics:** eGeMAPS extends the analysis to the frequency domain via Mel-Frequency Cepstral Coefficients (MFCCs) and formant frequencies (e.g., F_1, F_2). It also incorporates dynamic spectral descriptors like Spectral Flux, which captures the rate of change in the power spectrum. Unlike basic prosodic features, which describe how speech is modulated over time, these spectral features capture the resonant properties of the vocal tract influenced by articulatory configuration. Since distinct affective states often induce subtle changes in vocal tract shape (e.g., smiling vs. frowning), these spectral descriptors enable the model to disentangle acoustically similar yet semantically distinct emotions that prosody alone cannot separate.

- **Voice Quality & Timbre:** Beyond prosody, eGeMAPS includes micro-prosodic

features such as Jitter (frequency perturbation) and Shimmer (amplitude perturbation), which quantify voice "roughness" and "breathiness." These features are indispensable for distinguishing high-arousal negative emotions (e.g., the "harshness" of "Angry") from low-arousal ones (e.g., the "breathiness" of "Sad"), a nuance often lost in basic prosodic analysis.

However, directly ingesting all 88 eGeMAPS features into the LLM would result in excessive token consumption and potential noise. To resolve this, we employ a hybrid feature selection strategy to isolate the top $K = 20$ most discriminative attributes. This process integrates Lasso (L1) Regression [47] to induce sparsity and select features with high predictive value, alongside ANOVA [48, 49] to verify statistical significance between emotion classes.

These methods are chosen for their complementary strengths:

- **ANOVA (Filter Method):** Effectively identifies features that possess strong individual correlations with the target emotion, though it may overlook feature redundancy.
- **Lasso Regression (Embedded Method):** Assesses the joint contribution of variables, potentially retaining features that are weak individually but critical when combined (interaction effects).

The selected features from both methods are combined to ensure a comprehensive representation. Because ANOVA and Lasso frequently identify uniquely valuable descriptors alongside overlapping ones, taking the union of their respective selections yields a final, consolidated set of up to 30 distinct acoustic features. This data-driven approach ensures flexibility, allowing the system to dynamically adapt the feature set based on the specific acoustic characteristics of different datasets. The curated feature sets are detailed in Appendix A.

VAD Integration

Valence, Arousal and Dominance is a widely used dimensional representation of emotion in affective computing [17]. They are integrated into the feature set because discrete

emotion categories map directly onto this VAD space, rendering them highly salient features for classification. Valence measures the positivity (unpleasant to pleasant), Arousal indicates physiological intensity (calm to excited), and Dominance reflects the speaker's sense of control or power over the situation (submissive to dominant). Furthermore, our empirical analysis utilizing the aforementioned feature selection methods confirms this theoretical premise, demonstrating that VAD scores provide a high-level dimensional perspective that effectively complements the granular, low-level acoustic descriptors.

However, in real-world deployment, access to ground-truth VAD annotations are unavailable, and the system must rely on imperfect predicted scores. Models trained exclusively on perfect ground-truth VAD typically suffer significant performance degradation during inference due to this distribution shift. To mitigate this, we implement a Stochastic VAD Substitution strategy during training. For each VAD dimension, we apply a 50% probability of replacing the ground-truth score with its predicted counterpart. This exposes the network to the predicted state space during training, preventing over-reliance on ground-truth labels.

Additionally, human-annotated ground-truth VAD scores are inherently quantized (typically in increments of 0.5), whereas predicted scores represent a continuous distribution (e.g., increments of 0.1). To resolve this granularity mismatch and further regularize the model, we inject discrete uniform noise sampled from the set $\{-0.2, -0.1, 0.0, 0.1, 0.2\}$ into the ground-truth scores whenever they are selected. This noise injection smooths the discrete human annotations to match the continuous nature of the predictions, ensuring the LLM treats the VAD inputs as soft, guiding references rather than absolute, deterministic rules.

3.3.3 DFM

To enhance model robustness and prevent overfitting to specific acoustic modalities, we implement a DFM strategy. During training, each acoustic feature token is subject to a masking probability p, which increases linearly per epoch until it reaches a predefined

maximum threshold p_{max}. The probability for a given epoch is calculated as:

$$p = p_{max} \times \frac{\text{Current Epoch}}{\text{Total Epochs}}$$

This progressive regularization forces the model to perform inference with increasingly partial information, thereby mitigating over-reliance on any single dominant feature and improving adaptability to noisy, real-world inputs.

3.3.4 Prompt Optimization

The baseline methodology relies on generating natural language narratives to summarize speech features. While intuitive, this approach suffers from poor scalability: as the number of extracted features increases, the length of the descriptive narrative scales linearly. This introduces significant syntactic overhead, dilutes the information density and distracts the model's attention mechanism from critical acoustic data [50].

To resolve this, the proposed framework transitions from narrative descriptions to a Structured Prompt format (e.g., Key-Value pairs or JSON-like structures).

Figure 3.5: SpeechCueLLM Structured Prompt Example

Description (Omitted some features):

Target speech characteristics:
Valence:2.5/5
Arousal:3.0/5
Dominance:3.5/5

--- Pitch (Intonation) ---
Average Pitch: Medium
Pitch Stability (StdDev): Medium

--- Loudness (Energy) ---
Average Loudness: Medium
Loudness Variation (StdDev): Low

--- Rhythm & Speed ---
Speaking Rate: Very high

--- Voice Quality (Roughness/Breathiness) ---
Jitter (Voice Roughness): Medium
Shimmer (Voice Breathiness): High

--- Spectral Balance Related Features---
…
--- Formants Related Features---
…
--- MFCCs Related Features---
…

As illustrated in Figure 3.5, this format strips away linguistic redundancy, presenting the acoustic features in a raw, dense representation. Recent literature demonstrates that LLMs possess a robust capability to parse and reason over such structured data [51]. By organizing complex acoustic features into a concise, dictionary-like structure, we max-

imize information density and significantly reduce token consumption. This approach has been shown to yield performance comparable to, or exceeding, textual descriptions for complex tabular reasoning tasks while maintaining computational efficiency [52].

Recent research demonstrates that LLMs possess a robust capability to parse and reason over structured data [51]. By organizing complex acoustic features into a concise, dictionary-like structure, we maximize information density and reduce token consumption, which has been shown to yield comparable performance for complex data tasks [52].

Furthermore, we implement a Dynamic Prompt Construction mechanism to accommodate dataset-specific feature sets. Since the optimal acoustic descriptors are determined empirically for each corpus (e.g., via LASSO or ANOVA feature selection), a rigid template is unsuitable. Our framework dynamically generates the prompt schema at runtime, populating the structured input fields with only the specific feature keys relevant to the target dataset. This process includes a semantic mapping layer that translates raw OpenSMILE variable names into human-readable descriptors (e.g., mapping `F0semitoneFrom27.5Hz_sma3nz_amean` to "Average Pitch"). This step is essential to bridge the modality gap, ensuring that the technical acoustic features align with the LLM's pre-trained semantic understanding of vocal characteristics.

3.3.5 Dataset Scalability and Generalization

To overcome the limited taxonomy of the IEMOCAP-trained baseline (six emotions), we train two distinct model checkpoints to address different scopes of SER:

1. **Dyadic Model:** Optimized for dyadic interactions, excelling in scenarios requiring historical conversational context.
2. **Utterance-Level Fine-Grained Model:** Optimized for naturalistic, single-utterance scenarios, capable of identifying a broad spectrum of up to 16 fine-grained emotion classes (e.g., "Concerned", "Disappointed") often encountered in the wild.

Chapter 4

Experiments

This chapter details the rigorous experimental design and implementation strategies employed to evaluate the proposed fully automated Multimodal SER framework. Section 4.1 introduces the datasets utilized for benchmarking: the controlled, dyadic IEMOCAP corpus and the naturalistic, fine-grained MSP-Podcast corpus. It further provides a comprehensive statistical and dimensional analysis of these datasets to quantify the acoustic domain gap. Subsequently, Section 4.2 outlines the training setup, detailing the integration of the upstream pipeline modules (ASR, VAD Regressor, Gender Classifier) alongside the specific hyperparameter configurations for the LLM. Section 4.3 defines the evaluation metrics used to ensure unbiased performance assessment. Finally, Section 4.4 and Section 4.5 establish the comparative baseline models and delineate the methodology for the ablation studies, which systematically isolate the contributions of prompt structure, feature selection, and VAD integration.

4.1 Datasets

To rigorously evaluate the proposed fully automated pipeline, experiments are conducted across two fundamentally distinct affective corpora. While the foundational SpeechCueLLM [36] baseline utilized IEMOCAP [15] and MELD [53], we substitute MELD with the MSP-Podcast [16] corpus to better assess real-world applicability.

IEMOCAP is retained as a standard algorithmic benchmark to validate the framework's

capacity to process dyadic interactions and prototypical, high-intensity emotions. The substitution of MELD, however, is a critical methodological upgrade. Although MELD presents noisy, "in-the-wild" acoustic conditions, it remains fundamentally derived from scripted television dialogue and is restricted to coarse, primary emotion categories. In contrast, MSP-Podcast offers truly unscripted, spontaneous speech annotated with a highly granular emotional taxonomy (16 fine-grained classes). This substitution validates whether the model can successfully disentangle subtle, secondary affective states in authentic acoustic environments.

4.1.1 IEMOCAP

The Interactive Emotional Dyadic Motion Capture (IEMOCAP) corpus [15], collected by the Speech Analysis and Interpretation Laboratory (SAIL) at USC, stands as a premier benchmark for SER. The dataset comprises approximately 12 hours of audiovisual data featuring ten professional actors in five dyadic sessions, performing both scripted and improvised scenarios. While the corpus includes motion capture data, this research utilizes strictly the acoustic and textual modalities.

The dataset is distinguished by its high signal-to-noise ratio, resulting from controlled studio recording conditions. Furthermore, its dyadic structure provides rich conversational history, enabling models to leverage temporal context across dialogue turns—a critical factor for the proposed context-aware framework.

Data Partitioning

Session	1	2	3	4	5
Count	1365	1348	1533	1512	1622

Table 4.1: Utterance Distribution per Session in IEMOCAP

To ensure rigorous evaluation, we adhere to a speaker-independent protocol, isolating Session 5 as the held-out test set while using Sessions 1–4 for training. This strategy ensures that no speaker identity overlaps between training and testing, thereby preventing the model from overfitting to specific speaker characteristics. Table 4.1 confirms that the data is evenly distributed across sessions.

Class Selection and Preprocessing

Emotion	Count	Total Duration (min)
fru	1849	145.16
neu	1708	111.08
ang	1103	82.96
sad	1084	99.30
exc	1041	82.95
hap	595	43.05
sur	107	5.54
fea	40	1.82
oth	3	0.25
dis	2	0.08

Table 4.2: Emotion Distribution of IEMOCAP

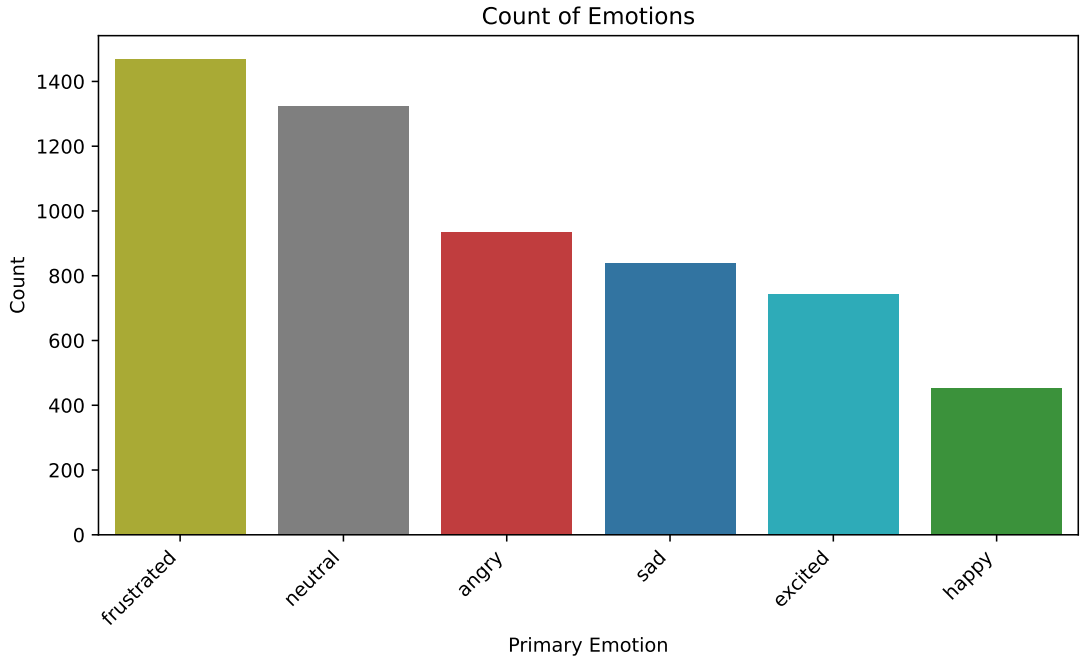

Figure 4.1: IEMOCAP Emotion Count

As shown in Table 4.2 and Figure 4.1, the raw dataset exhibits significant class imbalance. The classes for "Fear" and "Disgust" are severely underrepresented. Thus, we filter the dataset to focus on the four primary emotions plus "Excited" and "Frustrated", resulting in a final six-class taxonomy: Happy, Sad, Angry, Frustrated, Excited, and Neutral.

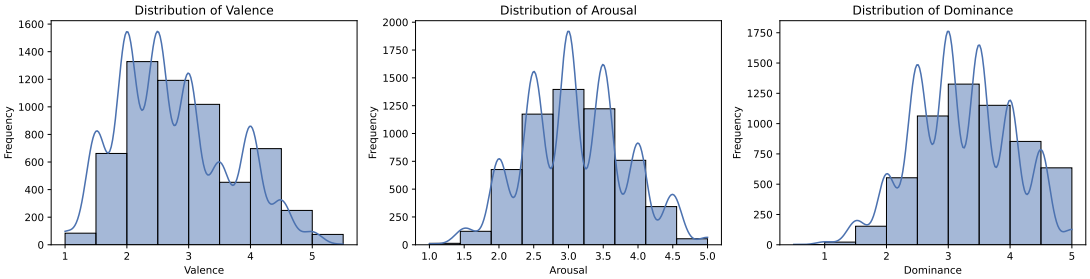

Figure 4.2: VAD Distribution of IEMOCAP

Statistical Characteristics

We further analyze the characteristics of the dataset:

- **VAD Distributions:** Valence exhibits a slight right skew, Arousal is largely symmetric, and Dominance shows a left skew. The mean values are 2.74, 3.09, and 3.21 (out of 5), respectively (Figure 4.2).
- **Demographics:** The dataset maintains a balanced gender ratio of 53% Male to 47% Female.
- **Length:** The average utterance duration is 4.60 seconds.

Valence-Arousal Space Analysis

To investigate the dimensional relationships between emotion classes, we visualize the interaction between Valence and Arousal in a 2D space, grounded in Russell's Circumplex Model [54].

Upon normalizing the IEMOCAP ratings to a scale of $[-1, 1]$ (centered at 0), distinct clustering patterns emerge as illustrated in Figure 4.3. The Valence dimension acts as the primary discriminator, clearly separating positive states ("Happy", "Excitement") which cluster in the right hemisphere, from negative states ("Angry", "Frustrated", "Sad") in the left.

The Arousal dimension facilitates distinction within these valence groups. On the positive spectrum, "Excited" exhibits consistently high activation (top-right quadrant), whereas "Happy" spans a wider range of arousal levels. Conversely, on the negative spectrum, "Sad" is characterized by low arousal (bottom-left quadrant), sharply con-

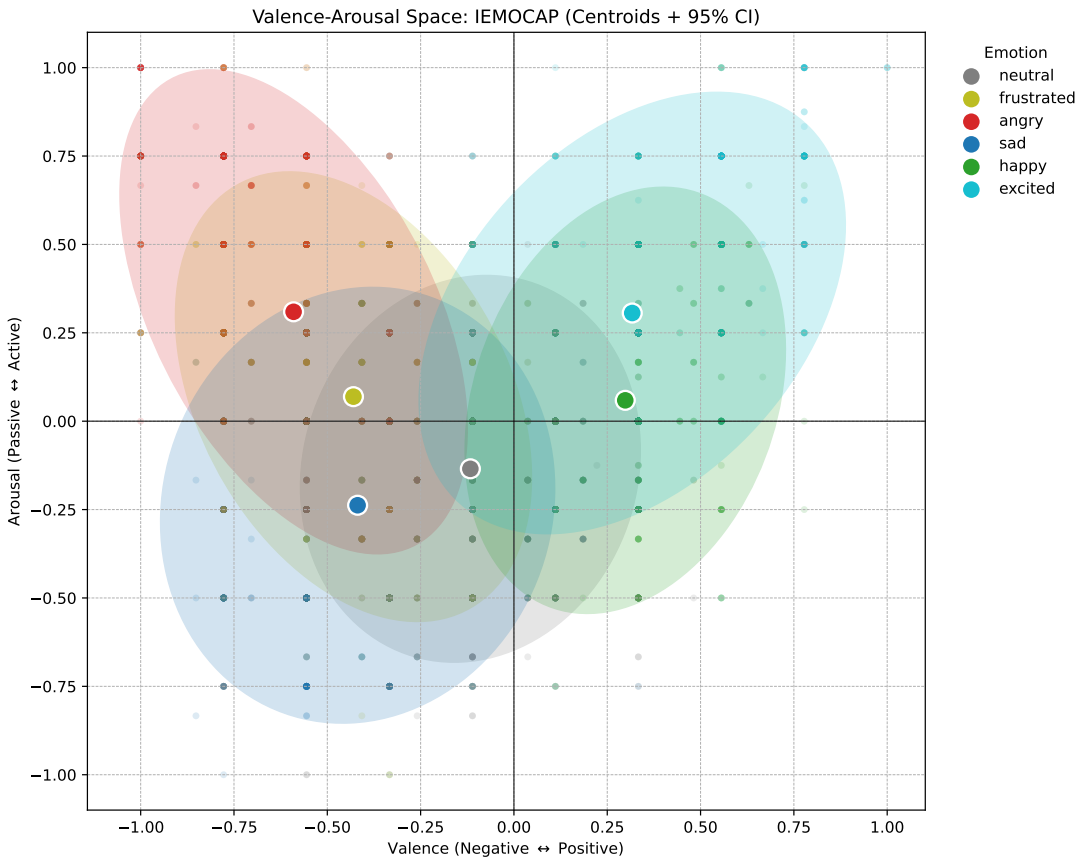

Figure 4.3: Valence-Arousal Space of IEMOCAP (Normalized Centroids)

trasting with "Anger", which occupies the high-arousal region. Notably, "Frustrated" mirrors the distributional variance of "Happy", spanning a broad range of arousal despite its negative valence. Finally, "Neutral" samples cluster near the origin of the valence axis but skew slightly toward lower arousal values.

4.1.2 MSP-Podcast

The MSP-Podcast corpus [16] represents a significant milestone in naturalistic SER research, culminating from over a decade of continuous data collection. The dataset comprises approximately 409 hours of audio sourced from diverse podcasts and audio-sharing platforms, featuring a vast and heterogeneous set of speakers. Unlike the acted dyadic interactions in IEMOCAP, MSP-Podcast captures spontaneous, real-world utterances.

Consequently, the emotional expressions within the corpus are often subtle and non-

prototypical, contrasting sharply with the exaggerated portrayals found in scripted datasets. The recordings inherently contain varying levels of background noise and acoustic overlapping, presenting a more rigorous challenge for acoustic modeling. Each segment is annotated by a minimum of five human raters, providing both categorical labels and dimensional VAD scores on a 7-point scale. This complexity positions MSP-Podcast as an essential benchmark for evaluating model robustness and generalization in "in-the-wild" deployment scenarios.

Emotion	**Count**	**Total Duration (min)**
neutral	39677	3569.86
concerned	21797	2146.83
happy	21336	2055.11
sad	11817	1123.84
angry	9588	922.57
frustrated	7552	754.05
amused	6823	640.44
annoyed	6572	648.81
disappointed	6213	589.38
excited	4471	439.03
surprise	4040	367.01
contempt	3745	362.37
confused	2159	195.11
disgust	2138	213.77
fear	697	64.38
depressed	505	49.17

Table 4.3: Emotion Distribution of MSP-Podcast

As illustrated in Table 4.3 and Figure 4.4, the dataset exhibits significant class imbalance, with a significant count of neutral samples.

Statistical Characteristics

Despite the categorical imbalance, the dimensional VAD distribution appears relatively symmetric. We summarize the key dataset statistics as follows:

- **VAD Distribution:** The dataset exhibits mean values of 3.91 for Valence, 4.30 for Arousal, and 4.45 for Dominance. This symmetric, neutral-leaning distribution reflects the subtle, non-exaggerated nature of naturalistic speech.
- **Demographics:** The dataset maintains a near-balanced gender ratio of 52% Male

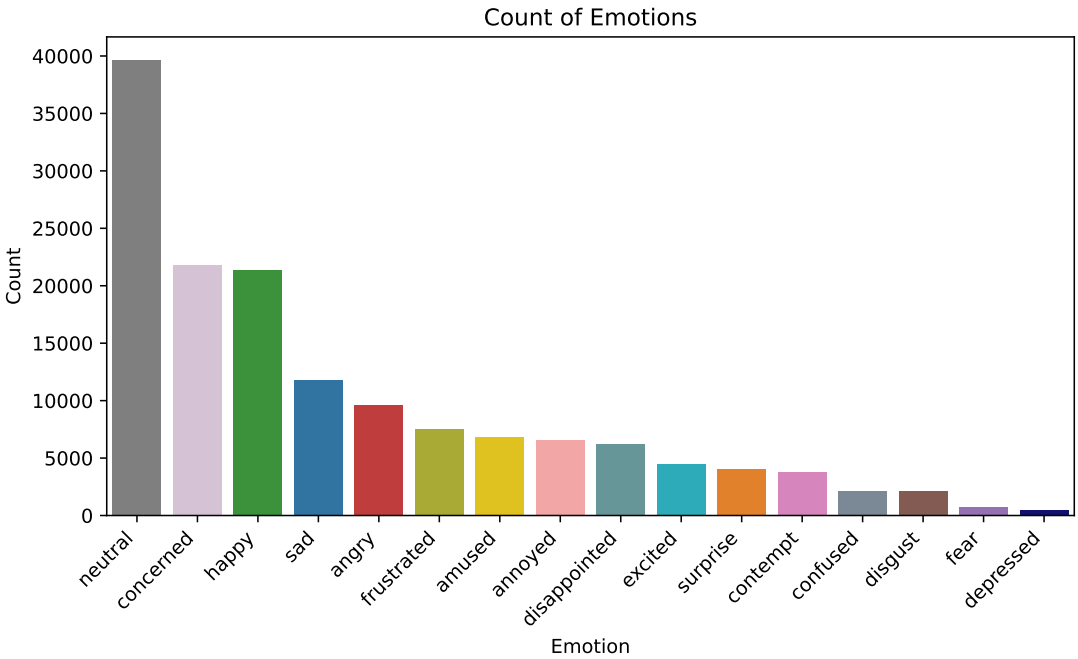

Figure 4.4: MSP-Podcast Emotion Count

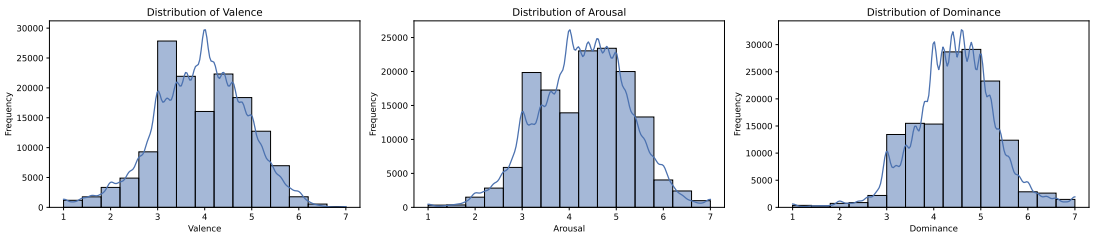

Figure 4.5: VAD Distribution of MSP-Podcast

to 48% Female, minimizing the risk of gender-based bias.

- **Length:** The average utterance duration is 5.69 seconds.

4.1.3 MSP-Podcast Subset

To mitigate the bias introduced by the severe class imbalance observed in the full corpus, we constructed a balanced subset. Specifically, we capped the duration of each emotion class to a maximum of one hour (approximately 500 utterances). This strategy ensures that the model treats all classes with equal importance during optimization, preventing overfitting to majority classes like “Neutral” or “Concerned”.

We then split the resulting balanced dataset into training and test sets using a test ratio

of 0.2. The split was performed via stratified subsampling based on emotion labels, ensuring that the test set preserves the same balanced class distribution as the training set.

Emotion	Count	Total Duration (min)
confused	538	59.66
surprise	529	60.06
neutral	527	59.99
fear	521	60.01
disappointed	513	60.01
happy	510	59.91
angry	509	60.16
amused	504	59.84
contempt	500	60.01
concerned	494	60.01
sad	493	60.01
excited	490	60.00
disgust	484	59.75
frustrated	477	59.81
annoyed	473	60.04
depressed	404	49.17

Table 4.4: Emotion Distribution of the MSP-Podcast Subset

As detailed in Table 4.4, this process yields a significantly more uniform distribution, with most categories containing between 450 and 550 samples. Notably, naturally rare classes such as "Depressed" remain slightly smaller due to data scarcity, but the disparity is drastically reduced compared to the original distribution.

Statistical Validation

To verify that this subsampling preserves the fundamental acoustic characteristics of the corpus, we compared the dimensional distributions of the subset against the full dataset.

- **Dimensional Consistency:** As shown in Figure 4.6, the subset preserves the original VAD distribution with remarkable fidelity. The average values for the subset are Valence: 3.82, Arousal: 4.37, and Dominance: 4.47. The fact that the distribution remains centered—even when "extreme" emotions are up-sampled to match "neutral" ones—supports the hypothesis that naturalistic emotional speech

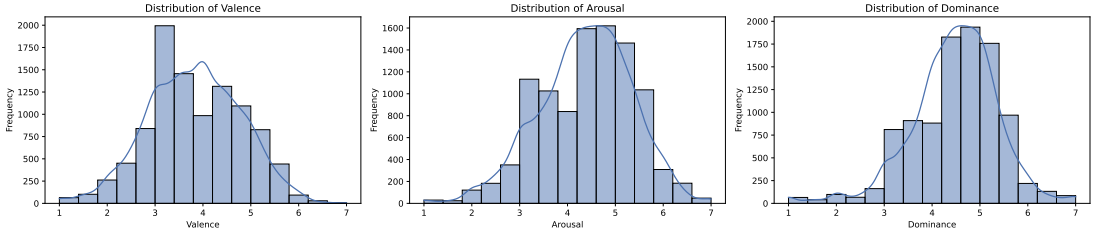

Figure 4.6: VAD Distribution of MSP-Podcast Subset

is less exaggerated and spectrally closer to the neutral position than acted speech.

- **Demographic Stability:** The subset maintains a gender ratio of 51% Male to 49% Female, preserving the demographic balance of the original corpus.
- **Attribute Preservation:** Table 4.5 compares the mean valence per emotion between the full dataset and the subset. The average deviation is negligible (≈ 0.06) for the majority of classes, indicating that the subset is statistically representative of the broader population.

Emotion	Mean (Original)	Mean (Subset)
angry	2.8483	2.7925
annoyed	3.1819	3.1295
frustrated	3.1943	3.1879
disgust	3.3086	3.3025
disappointed	3.3561	3.3571
sad	3.3907	3.3894
contempt	3.5978	3.5615
fear	3.6575	3.6506
concerned	3.7953	3.7349
confused	3.9176	3.9139
neutral	3.9575	3.9708
depressed	3.9642	3.9663
surprise	4.4870	4.4965
amused	4.7855	4.7748
happy	4.8361	4.8269
excited	4.8991	4.9083

Table 4.5: Comparison of Mean Valence between Original Dataset and Subset

For the remainder of this report, all references to "MSP-Podcast" denote this balanced subset unless explicitly stated otherwise.

Valence-Arousal Space Analysis

We apply the same dimensional analysis to the MSP-Podcast subset, normalizing the original 7-point scale to a range of $[-1, 1]$. To facilitate a direct and meaningful comparison with IEMOCAP—and to maintain visual clarity—we restrict this visualization to the standard primary emotions (Happy, Sad, Angry, Fear, Disgust, Surprise, Neutral) alongside the two secondary emotions present in the IEMOCAP taxonomy (Excited, Frustrated).

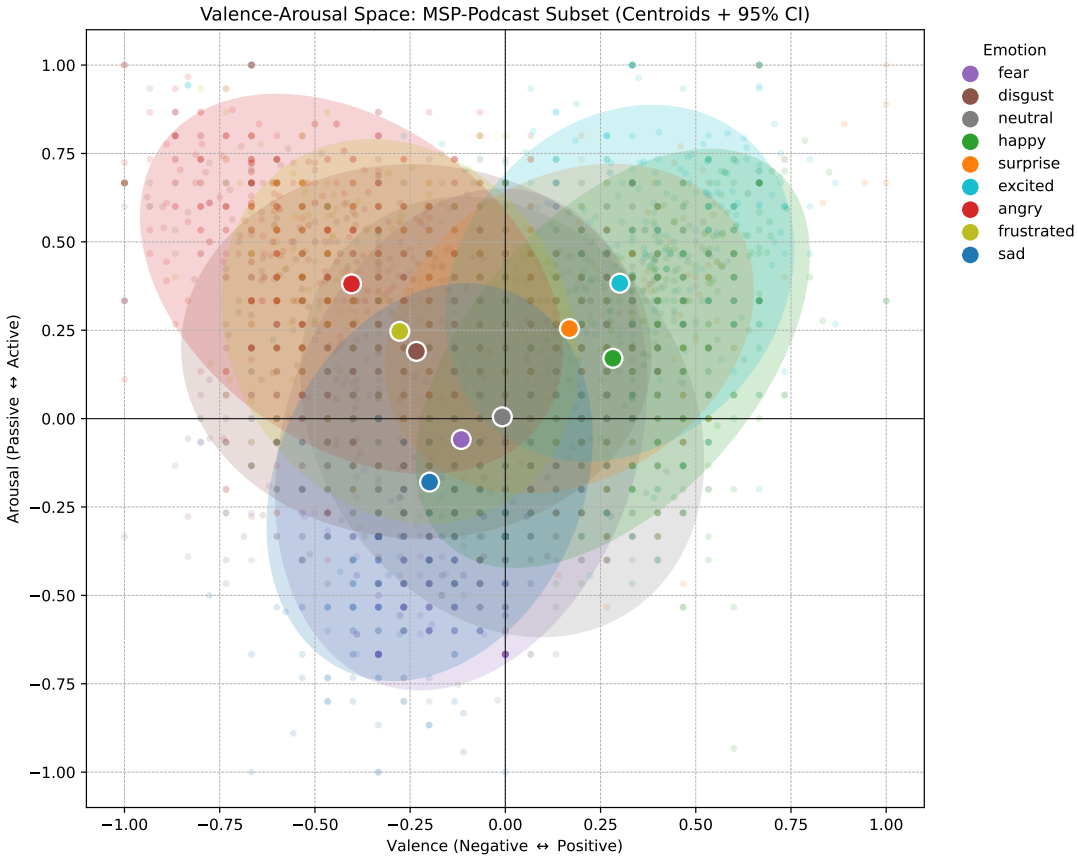

Figure 4.7: Valence-Arousal Space of MSP-Podcast Subset

As illustrated in Figure 4.7, the overall distribution exhibits structural similarities to that of IEMOCAP (Figure 4.3). However, specific nuances emerge in the naturalistic domain:

- **Negative Valence Clustering:** Both "Disgust" and "Fear" reside in the negative valence region, consistent with theoretical expectations.
- **Arousal Distinctions:** A divergence is observed in arousal levels. "Disgust"

tends toward active arousal, whereas “Fear” exhibits surprisingly passive arousal in this dataset. In fact, the centroid for “Fear” is located in close proximity to the origin, making it the emotion nearest to “Neutral” in terms of Euclidean distance within the VA space. This likely reflects the subtle nature of “Fear” in podcasts (e.g., worry/concern) rather than acute panic.

4.1.4 Datasets Comparison

To bridge the domain gap between acted and naturalistic speech, we performed a comparative analysis of the emotional centroids of both datasets within the Valence-Arousal (VA) space.

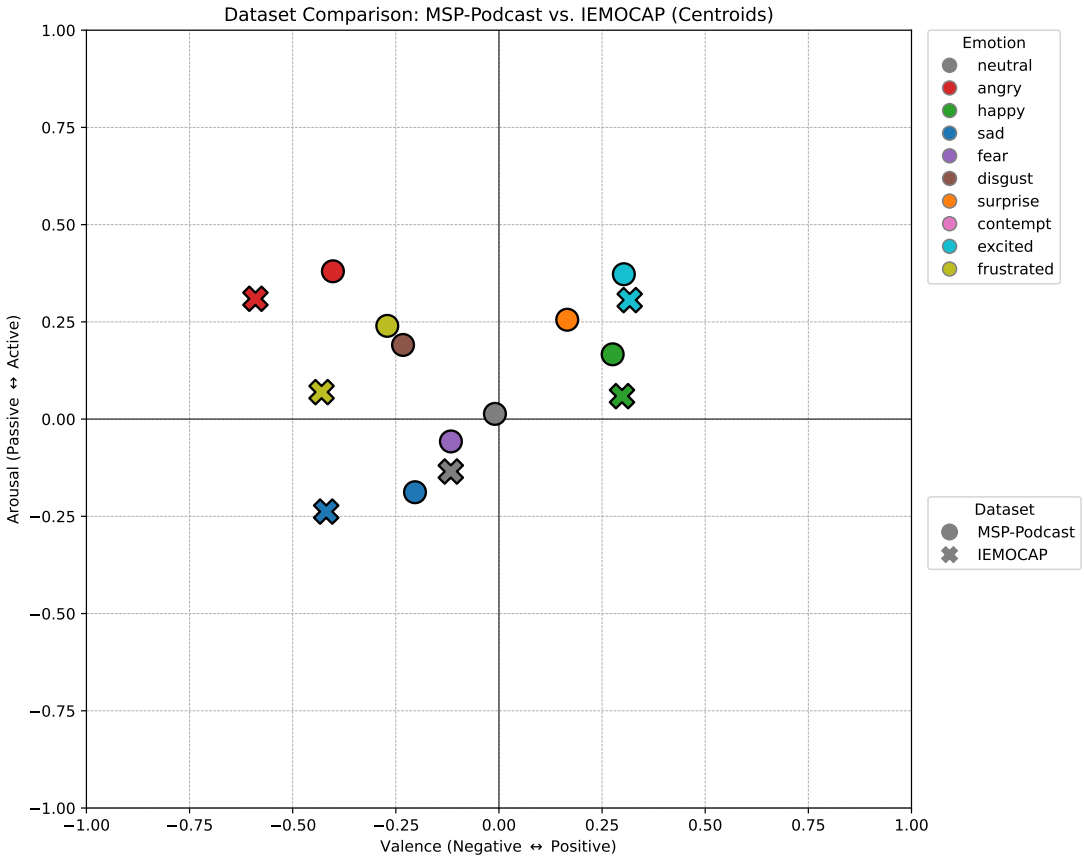

Figure 4.8: Dataset Comparison: IEMOCAP VS MSP-Podcast

As illustrated in Figure 4.8, distinct distributional shifts are observable between the two domains:

- **Negative Valence Extremity:** In the negative valence spectrum, IEMOCAP

exhibits significantly more extreme values compared to MSP-Podcast. This phenomenon suggests that professional actors tend to express negative states (e.g., "Angry", "Sad") with greater intensity and prototypical clarity than what is typically observed in spontaneous conversation. Conversely, in the positive valence spectrum, both datasets converge, with "Excited" and "Happy" showing comparable average valence.

- **Arousal Baseline Shift:** A systematic shift is observed in the arousal dimension. Contrary to the expectation of "high-energy" acting, IEMOCAP samples demonstrate a generally lower average arousal compared to their MSP-Podcast counterparts. This may reflect the acoustic reality of podcasts, where background noise, compression, and varying recording conditions artificially inflate energy features, or simply that naturalistic conversations involve more dynamic engagement than controlled studio scripts.
- **Inter-Class Separability vs. Ambiguity:** A critical advantage of IEMOCAP is the wide spatial distribution of its centroids. The clear separation between emotion clusters reduces decision boundary ambiguity, simplifying the classification task. In contrast, MSP-Podcast emotions cluster tightly around the neutral origin. This "central tendency" reflects the subtle nature of real-world emotion, introducing significant ambiguity and complexity to the recognition task.

Class-Wise Dimensional Analysis

To further examine the internal consistency of emotion classes, we analyze the distribution of VA points for each emotion category individually.

Comparative Analysis: These two distributions reveals a fundamental divergence between acted and naturalistic affective spaces.

As observed in Figure 4.9, the IEMOCAP distribution appears highly polarized and discrete. This is most evident in negative emotions such as "Angry", "Sad", and "Frustrated", where a substantial volume of data points are concentrated beyond the -0.5 valence threshold. This polarization serves as quantitative evidence of theatrical exaggeration; actors in a studio environment tend to project emotions with high magnitude

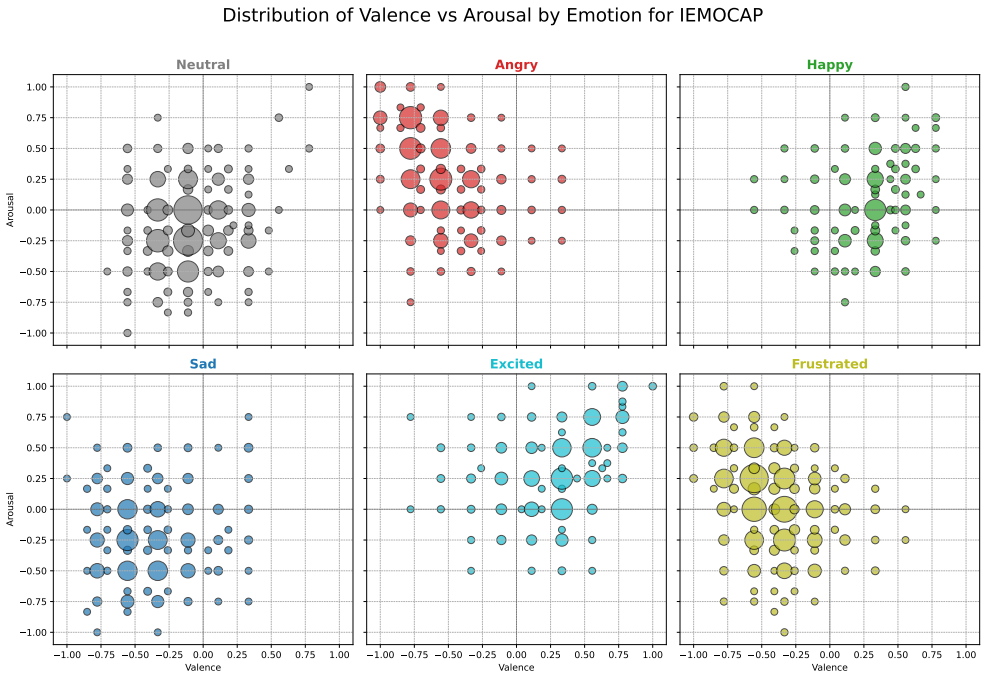

Figure 4.9: Valence-Arousal Space of IEMOCAP by Emotions

to ensure clarity. While this distinct separation simplifies the classification task, it suggests that the primary challenge in IEMOCAP is not ambiguity, but rather the potential for the model to overfit to these prototypical, high-intensity expressions.

In stark contrast, the dimensional space of the naturalistic MSP-Podcast corpus (Figure 4.10) is more continuous. While the class centroids align with expected quadrants, we observe significant intra-class variance and spectral overlap. Outliers frequently appear in opposing quadrants (e.g., "Happy" samples with negative valence due to sarcasm or context), creating fuzzy decision boundaries.

The lack of clear separation in MSP-Podcast compared to IEMOCAP introduces irreducible ambiguity, confirming that dimensional coordinates—while powerful—are insufficient as standalone discriminators for naturalistic speech and must be augmented by contextual cues.

4.1.5 Summary of Dataset Analysis

The comparative analysis of IEMOCAP and MSP-Podcast highlights a fundamental trade-off in SER research: Control versus Realism.

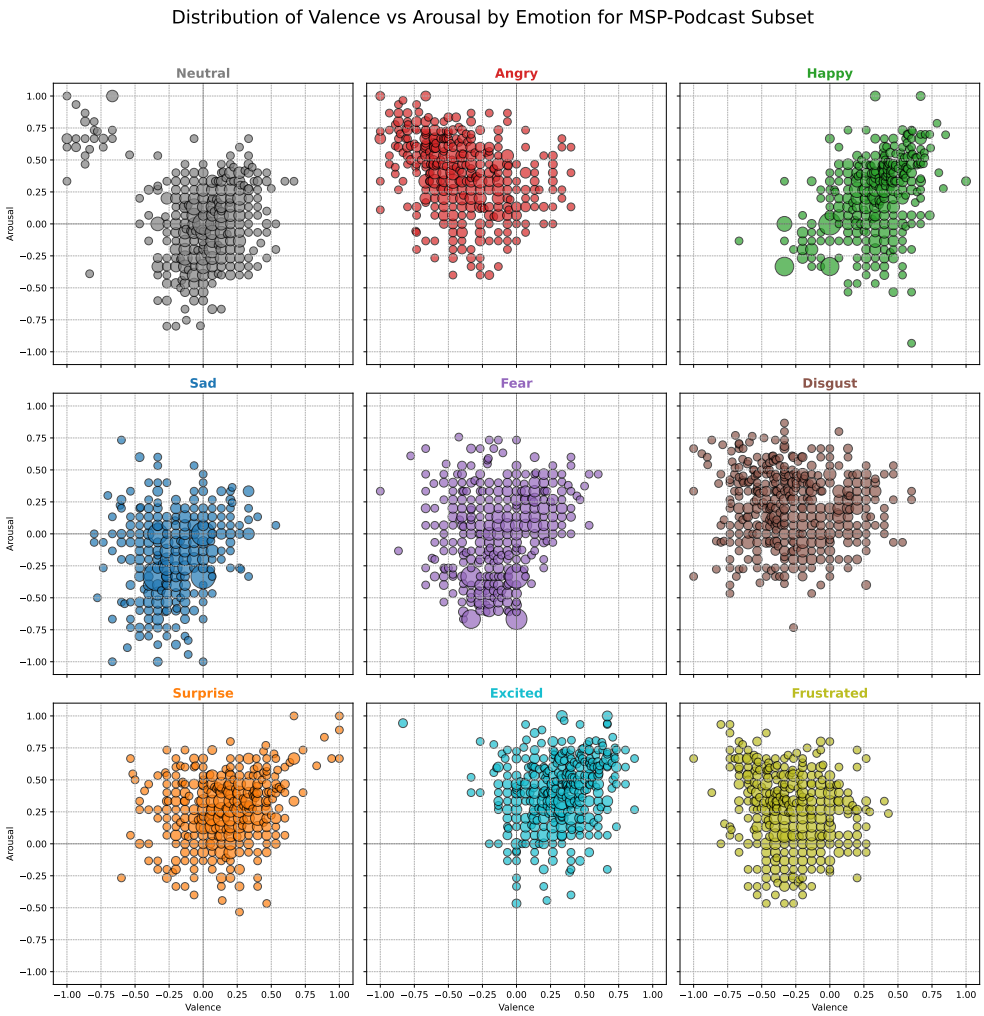

Figure 4.10: Valence-Arousal Space of MSP-Podcast Subset by Emotions

- **IEMOCAP (Control):** This dataset provides a controlled environment with high-fidelity, noise-free audio and rich dyadic conversational context. It serves as a standard benchmark for algorithmic validation. However, it is limited by restricted speaker diversity (only 10 actors) and a susceptibility to theatrical exaggeration, potentially yielding models that fail to generalize to subtle, low-arousal cues.

- **MSP-Podcast (Realism):** In contrast, MSP-Podcast captures the complexity of spontaneous human communication, sourced from diverse "in-the-wild" environments. While it accurately reflects naturalistic speech patterns, it introduces significant challenges including varying signal quality, background noise, and acoustic overlapping, alongside high semantic ambiguity in the decision bound-

aries.

By evaluating on both domains, this research aims to validate that the proposed LLM-based framework can transcend these boundaries—leveraging the clarity of acted speech for learning basic representations while adapting to the noise and nuance of naturalistic environments.

4.2 Training Setup

4.2.1 Implementation Details

The proposed framework is implemented using the PyTorch framework, accelerated via DeepSpeed to optimize memory efficiency on a single *Tesla V100-PCIE-32GB* GPU.

The preprocessing pipeline integrates several specialized models:

- **Speech Recognition:** We utilize the *Whisper v3 large-turbo* model [55] for Automatic Speech Recognition (ASR). This model is selected for its optimal balance between transcription accuracy and computational throughput. Its transcription reliability is evaluated using Word Error Rate (WER).
- **VAD Regression:** Continuous VAD scores are predicted using the fine-tuned model from [56], which is specifically optimized for paralinguistic dimensional estimation. The regressor's performance is measured using the Concordance Correlation Coefficient (CCC) to assess the alignment between predicted and ground-truth continuous traces.
- **Gender Classification:** We adapted and fine-tuned a pre-trained Wav2Vec 2.0 (W2V2) model [27] to perform binary speaker gender classification, evaluated using standard Weighted Accuracy.
- **Feature Extraction:** The openSMILE toolkit [46] is employed to extract the eGeMAPS [45].

4.2.2 Hyperparameter Configuration

Table 4.6 summarizes the training hyperparameters. Consistent with the baseline, we employ LoRA [57] for parameter-efficient fine-tuning. By freezing the pretrained LLM weights and injecting trainable rank decomposition matrices, LoRA significantly reduces computational overhead while maintaining model performance.

Hyperparameter	**Baseline**	**Proposed**	
	IEMOCAP	**IEMOCAP**	**MSP-Podcast**
LoRA Rank (r)	16	16	32
LoRA Scaling Factor (α)	16	16	32
Gradient Accum. Steps	8	8	8
Effective Batch Size	8	8	8
Learning Rate	3×10^{-4}	1×10^{-4}	1×10^{-5}
Optimizer	FusedAdam	FusedAdam	FusedAdam
β_1	0.9	0.9	0.9
β_2	0.95	0.95	0.95
Learning Rate Schedule	Linear	Linear	Linear
Warmup Ratio	0.1	0.1	0.1
Epochs	15	15	15
Early Stopping	Yes	Yes	Yes
Max Token Length	2500	2500	2500
Max Masking Prob (p_{max})	-	-	0.1
Context Window Size (N)	8	8	0

Table 4.6: Comparative Training Hyperparameters for the Baseline, Dyadic (IEMOCAP), and Fine-Grained (MSP-Podcast) Models

Hyperparameter Configuration and Baseline Consistency

To establish a rigorous comparative analysis, foundational training configurations were kept strictly consistent with the SpeechCueLLM baseline where possible. Both the Dyadic (IEMOCAP) and Fine-Grained (MSP-Podcast) models were trained for 15 epochs utilizing DeepSpeed's FusedAdam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.95$) with a linear learning rate decay schedule and a 10% warmup phase. An effective batch size of 8 was universally maintained to ensure stable weight updates. Furthermore, a maximum context window of 2500 tokens and an early stopping mechanism were applied universally to maximize historical context utilization while preventing overfitting. However, to accommodate the eGeMAPS and the distinct acoustic complexities of the target

corpora, specific hyperparameter deviations from the baseline were necessary:

Context Window Size (N): To effectively capture the temporal dynamics of dyadic interactions, the conversational context window for the IEMOCAP model was set to $N = 8$ historical utterances. While the original SpeechCueLLM study utilized $N = 12$, computational constraints necessitated this reduction. To ensure a strictly rigorous and fair comparative analysis, both the baseline model and the proposed Dyadic model was trained and evaluated under this identical $N = 8$ configuration. Conversely, for the MSP-Podcast dataset, the context window was disabled ($N = 0$). Because this corpus consists of independent, isolated speech segments rather than continuous multi-turn dialogues, sequential historical context is structurally inapplicable. Consequently, the model must rely exclusively on the target utterance itself to decode the spontaneous paralinguistic cues.

LoRA Dimension (r): For the Dyadic model, we maintain the baseline's rank of $r = 16$ ($\alpha = 16$). Conversely, for the naturalistic MSP-Podcast dataset, the rank is increased to $r = 32$. This expansion provides the adapter modules with greater representational capacity, a critical adjustment for capturing the complex, non-linear dependencies inherent to spontaneous, "in-the-wild" affective expressions.

Learning Rate: Empirical observations indicated that the baseline's learning rate of 3×10^{-4} was overly aggressive for our high-dimensional acoustic feature space. Consequently, the learning rate for the Dyadic model was targetedly reduced to 1×10^{-4} to ensure stable gradient convergence. For the Fine-Grained model, where the optimization landscape is significantly noisier due to severe acoustic variability, the rate was further decelerated to 1×10^{-5} to guarantee stable optimization.

DFM Implementation: DFM is applied exclusively to the Fine-Grained Model, as the MSP-Podcast dataset features naturalistic audio and is trained with a highly stable, lower learning rate, DFM effectively acts as a strong regularizer, forcing the network to distribute its attention across features and preventing over-reliance on any single modality. Conversely, the Dyadic Model is trained on clean, studio-recorded audio using a comparatively higher learning rate. In this regime, the optimization process already exhibits sufficient inherent stochasticity; introducing the additional noise of

DFM compounded this instability, leading to degraded performance, and was therefore omitted.

4.3 Evaluation Metrics

To ensure rigorous benchmarking, we employ three primary performance metrics: Unweighted Accuracy (UA), Weighted Accuracy (WA), and Macro-F1 Score (MF1). Let C denote the number of emotion classes, N the total number of samples, and N_i the number of samples in class i.

Weighted Accuracy (WA)

Weighted Accuracy, commonly referred to as standard accuracy, measures the overall ratio of correct predictions across the entire dataset:

$$WA = \frac{\sum_{i=1}^{C} TP_i}{N} \tag{4.1}$$

where TP_i represents the number of True Positives for class i. While intuitive, WA is biased towards majority classes in imbalanced datasets; however, we report it to maintain comparability with baseline studies that utilize this standard.

Unweighted Accuracy (UA)

Unweighted Accuracy is defined as the arithmetic mean of the recall scores per class:

$$UA = \frac{1}{C} \sum_{i=1}^{C} \text{Recall}_i = \frac{1}{C} \sum_{i=1}^{C} \frac{TP_i}{N_i} \tag{4.2}$$

By weighting each class equally regardless of its sample size, UA provides an unbiased estimate of model performance. This metric is particularly critical for evaluating the MSP-Podcast dataset, ensuring that minority classes (e.g., “Sad”) are not overshadowed by majority classes (e.g., “Neutral”).

Macro-F1 Score (MF1)

The F1-score is the harmonic mean of precision (P_i) and recall (R_i) for a specific class i, providing a balanced measure that penalizes extreme disparities between the two:

$$F1_i = 2 \cdot \frac{P_i \cdot R_i}{P_i + R_i} \tag{4.3}$$

The Macro-F1 score is subsequently calculated as the unweighted average of these class-wise F1 scores:

$$MF1 = \frac{1}{C} \sum_{i=1}^{C} F1_i \tag{4.4}$$

MF1 serves as a robust metric for multi-class classification, offering a comprehensive assessment of the model's ability to minimize both False Positives and False Negatives across all emotional categories.

4.4 Baseline & SOTA Models

To rigorously assess the efficacy of the proposed framework, we benchmark performance against both the established SpeechCueLLM baseline and a diverse set of state-of-the-art (SOTA) models. These comparisons are categorized into two distinct architectural paradigms: Conventional Non-LLM SOTA benchmarks and Multimodal LLM SOTA benchmarks.

4.4.1 Non-LLM SOTAs

These models represent the current SOTA in pure audio representation learning. They function by encoding raw audio waveforms into dense vector representations, which are subsequently used for emotion classification without textual or semantic conditioning.

- **WavLM** [58]: A robust large-scale model pre-trained on massive unlabeled speech data, optimized for diverse downstream speech tasks including SER.
- **emotion2vec** [13]: A specialized model fine-tuned specifically for emotion recognition tasks using self-supervised learning on emotional speech data.

4.4.2 Multimodal LLM SOTAs

To evaluate the capabilities of general-purpose foundation models in the SER domain, we conduct zero-shot evaluations on several leading LLMs. For these experiments, the conversation history and target utterance transcription are consistently provided as textual context. Acoustic information is integrated either through discrete textual descriptors or by directly processing the raw audio via an internal encoder:

- **Text-Only LLMs:** For *LLaMA-3-8B-Instruct* [59], we adhere to the baseline methodology by including the acoustic feature descriptors as part of the textual context.
- **Multimodal LLMs:** For *Qwen2-Audio-7B-Instruct* [41] and *Qwen3-Omni-30B-A3B-Instruct* [42], we replace the textual feature descriptors with the raw audio path of the target utterance. This allows the model to leverage its internal audio encoder to extract paralinguistic cues directly from the signal while retaining the shared semantic context.
- **Audio-Captioning LLMs:** While operating on the raw audio similarly to the multimodal models, *Qwen3-Omni-30B-A3B-Captioner* [42] and *DeSTA2.5-Audio* [60] possess the dual capability of generative description and direct classification. Therefore, we evaluate two distinct prompting strategies:
  - **Direct Tagging:** Instructing the model to output a specific emotion label.
  - **Caption Generation:** Instructing the model to describe the audio, followed by a keyword filtering process to map the generated description to an emotion category.

  For each model, we report the higher score obtained between these two strategies to ensure a fair assessment of its maximum capability.

4.5 Ablation Studies

To isolate the distinct contributions of the proposed enhancements, this section systematically evaluates each component of our framework. We analyze the impact of prompt structuring, feature selection, and the integration of VAD scores across the fine-tuned configurations for both the Dyadic and Fine-Grained models.

4.5.1 Prompt Structure Analysis

We hypothesize that a structured prompt—presenting speech features in a formalized tabular or key-value format—significantly improves the LLM's ability to parse and attend to acoustic information compared to dense natural language descriptions. To validate this, we compare two prompt configurations:

- $P_{unstruct}$ **(Unstructured):** A naive baseline configuration where transcriptions and acoustic features are concatenated into a single, continuous paragraph without clear headers or delimiters.
- P_{struct} **(Structured):** The proposed methodology, utilizing a structured, table-like format to organize speech characteristics, designed to maximize information density and retrieval efficiency.

Crucially, this comparative analysis is conducted exclusively using the baseline prosodic feature set ($N = 5$). Evaluating the unstructured narrative format on our extended feature set (which contains up to 30 acoustic descriptors) would exponentially increase token consumption, inevitably leading to severe prompt bloat and attention dilution. By restricting this ablation to the low-dimensional baseline features, we ensure a fair, isolated evaluation of the prompt schema itself, preventing context-window degradation from confounding the results.

4.5.2 Feature Selection

Building upon the established efficacy of the structured prompt format, we next investigate the optimal strategy for acoustic feature selection. To ensure a controlled

environment that isolates the impact of the chosen features, all configurations in this experiment are trained utilizing the Stochastic VAD Substitution strategy. To ensure ecological validity, it is critical to note that during inference, all VAD-inclusive configurations rely strictly on predicted VAD scores. This constraint accurately simulates real-world deployment where perfect annotations are unavailable. To identify the most discriminative subset from the eGeMAPS, we compare four distinct selection methodologies, utilizing a randomly selected feature set as a control.

- **Continuous Statistical Selection (Numeric):** This standard empirical method applies LASSO regression and ANOVA directly to the gender-standardized, continuous numeric eGeMAPS features (as detailed in the Methodology section) to isolate the most statistically significant acoustic descriptors.
- **Discretized Statistical Selection (Categorical):** To evaluate whether the LLM processes bucketed concepts more effectively than raw continuous distributions, this approach first converts the gender-standardized numeric features into ordinal categorical bins prior to applying the LASSO and ANOVA selection algorithms.
- **LLM-Guided Semantic Selection:** Inspired by the LLM-Select framework [61], this approach bypasses traditional statistical algorithms to leverage the foundation model's intrinsic semantic knowledge. The LLM is provided with the comprehensive list of available features alongside the task context and is prompted to rank the descriptors based on their theoretical importance for emotion classification.
- **Random Selection (Control):** A setting where 30 acoustic features are selected entirely at random. This serves as a control to validate whether the empirical and semantic selection methods provide a meaningful advantage over arbitrary features.

4.5.3 VAD Integration Analysis

Having established the continuous statistical (numeric) approach as the optimal feature selection methodology, we isolate the impact of VAD integration and our proposed

training regimen.

We evaluate three distinct settings:

- **No VAD:** The model relies entirely on the selected acoustic features, providing a for performance without any dimensional emotional context.
- **Ground-Truth (GT) Only Training:** The model is trained exclusively on GT VAD annotations but evaluated using predicted VAD. This setting is designed to highlight the model's vulnerability to distribution shift and performance degradation when transitioning from perfect training data to noisy, real-world inputs.
- **Stochastic VAD Substitution (Proposed):** The model is trained using our proposed 50% mixed-probability regimen with discrete noise injection. This configuration evaluates the efficacy of our regularization strategy in bridging the gap between training and inference conditions.

Chapter 5

Result Analysis

This chapter presents a comprehensive empirical evaluation of the framework. Section 5.1 begins with a comparative performance analysis against established SOTAs on both the acted IEMOCAP and naturalistic MSP-Podcast datasets, followed by a granular, class-wise examination of the framework's efficacy in disentangling complex affective states. Subsequently, Section 5.2 investigates the model's cross-corpus generalization capabilities to validate the robustness of the learned paralinguistic representations. To isolate the specific contributions of the architectural components, Section 5.3 detail the ablation studies, quantifying the impact of the Structured Key-Value Prompting strategy, feature selection methodologies, and VAD integration strategies. Building upon this, Section 5.4 conducts an Upper Bound analysis to establish the theoretical upper bound of the framework by evaluating the model with perfect VAD scores, investigating the system's susceptibility to distribution shifts in VAD scores. Section 5.5 and Section 5.6 rigorously assess the framework's resilience to downstream error propagation by introducing realistic transcription and gender misclassification noise from the upstream auxiliary modules. Finally, Section 5.7 concludes with a qualitative analysis, illustrating the LLM's multimodal reasoning process through specific prompt examples and prediction outputs.

5.1 Results

To rigorously assess the efficacy of the proposed fully automated framework, we benchmark its performance against a diverse suite of SOTA models. These benchmark models encompass both specialized unimodal audio architectures and large-scale multimodal LLMs. The comparative evaluation is conducted across the acted IEMOCAP and naturalistic MSP-Podcast datasets, utilizing Unweighted Accuracy (UA), Weighted Accuracy (WA), and Macro-F1 (MF1) to ensure a balanced assessment across skewed class distributions.

5.1.1 Performance Analysis

Table 5.1: Main Performance Comparison.(%) GT transcription and speaker gender are utilized during inference for all LLMs.

	IEMOCAP			**MSP-Podcast**		
Model Configuration	**UA ↑**	**WA ↑**	**MF1 ↑**	**UA ↑**	**WA ↑**	**MF1 ↑**
Finetuned Non-LLM						
Emotion2Vec	63.33	67.94	62.26	17.58	17.95	14.45
WavLM	49.29	52.40	47.50	17.25	17.45	12.91
Pretrained LLMs						
Pretrained Llama-3-8B	48.54	46.89	45.52	10.92	11.23	8.43
Qwen2-Audio	46.14	41.19	41.95	12.22	12.61	8.67
Qwen3-Omni	58.78	57.28	54.21	18.17	20.72	18.18
Qwen3-Omni Captioner	55.00	53.60	51.48	16.46	16.96	15.42
DeSTA-2.5	53.69	46.58	45.85	20.00	20.27	18.65
Finetuned LLMs						
SpeechCueLLM	70.63	71.97	70.66	18.97	19.26	18.57
Ours	**71.36**	**73.67**	**71.99**	**26.55**	**26.98**	**25.13**

As detailed in Table 5.1, the proposed framework not only surpasses the SpeechCueLLM baseline, but also outperforms both Non-LLM models and LLM competitors across both datasets.

- **The Backbone Delta (Impact of Fine-Tuning):** A critical observation is the massive performance gap between the zero-shot Pretrained Llama-3-8B backbone and our proposed framework. On IEMOCAP, the Macro-F1 leaps from 45.52%

to 71.99%, and on MSP-Podcast, it surges from a mere 8.43% to 25.13%. This stark contrast empirically proves that while foundation models possess strong semantic reasoning, they cannot natively resolve paralinguistic emotion without specialized acoustic representations and targeted fine-tuning.

- **Superiority over Native Multimodal models:** Our framework also outperforms LALMs and native multimodal foundation models, despite those models processing continuous raw audio directly. Notably, Qwen3-Omni achieves a respectable 54.0% MF1 on IEMOCAP and 19.0% on MSP-Podcast, yet falls significantly short of our architecture. This validates that extracting and structuring dense, domain-specific acoustic features (eGeMAPS) is more effective for fine-grained emotion recognition than relying on general-purpose audio encoders.
- **Improvement over SpeechCueLLM (IEMOCAP):** Improvement is present but modest, with a 1.33% increase in Macro-F1 score. This marginal gain is attributed to the dominance of conversational context in dyadic scenarios. The long-term dialogue history provides strong cues that often overshadow acoustic subtleties, making the model less reliant on the enhanced feature set. Nevertheless, the consistent gain confirms that the proposed mechanism contributes additive value even in context-rich environments.
- **Improvement over SpeechCueLLM (MSP-Podcast):** We observe a substantial performance leap, with Macro-F1 improving by over 6.56% compared to SpeechCueLLM. This validates the efficacy of our expanded feature set and structured prompting strategy in capturing the nuances of "in-the-wild" speech where linguistic content alone is often ambiguous.

5.1.2 Class-wise F1 Scores

IEMOCAP Analysis

As presented in Table 5.2, the proposed framework exhibits a distinct performance trade-off characterized by substantial improvements in high-arousal states. Most notably, the recognition of "Excited" and "Angry" sees dramatic increases of +7.7% and +6.2%,

Table 5.2: Class-wise F1-score analysis on IEMOCAP (%)

Emotion Class	**Baseline ↑**	**Ours ↑**	**Gain (Δ)**
Neutral	74.2	**76.5**	+2.3
Happiness	**60.4**	55.3	-5.1
Sadness	**83.9**	82.2	-1.7
Anger	63.7	**69.9**	+6.2
Excited	70.4	**77.7**	+7.7
Frustrated	**71.3**	70.3	-1.0
MF1	*70.7*	***72.0***	*+1.3*

respectively, alongside a steady gain in the "Neutral" class (+2.3%). This suggests that the inclusion of extended acoustic features—particularly spectral dynamics and precise intensity metrics—is highly effective at resolving the energetic signatures of active emotions.

Conversely, these gains are accompanied by minor regressions in "Sadness" (-1.7%) and "Frustrated" (-1.0%), as well as a more pronounced degradation in "Happy" (-5.1%). The sharp decline in "Happy" can be primarily attributed to severe class imbalance; its significantly lower sample count within the dataset leaves it highly susceptible to being overshadowed in the feature space, often resulting in misclassification into the acoustically similar but more dominant "Excited" category. Ultimately, however, the targeted enhancements in resolving complex, high-energy states drive an overall improvement in the framework's Macro-F1 score (+1.3%).

MSP-Podcast Analysis

As detailed in Table 5.3, the proposed framework consistently outperforms the baseline, achieving a +6.5% absolute increase in the Macro F1-score. The model demonstrates exceptional efficacy in disentangling acoustically distinct emotions, with the most substantial gains observed in "Angry" (+17.8%), "Happy" (+16.6%), and "Disappointed" (+13.1%). This validates the hypothesis that the inclusion of proposed feature set provides the necessary acoustic resolution to accurately distinguish a wide array of both primary and secondary emotional states.

However, this overall performance boost entails a specific trade-off regarding extremely

Table 5.3: Class-wise F1-score analysis on MSP-Podcast (%)

Emotion Class	**Baseline ↑**	**Ours ↑**	**Gain (Δ)**
Primary Emotions			
Neutral	18.5	**21.6**	+3.1
Happiness	15.3	**31.9**	+16.6
Sadness	17.4	**26.9**	+9.5
Anger	18.6	**36.4**	+17.8
Fear	21.4	**27.6**	+6.2
Disgust	26.1	**31.4**	+5.3
Surprised	23.0	**23.8**	+0.8
Contempt	**8.0**	1.5	-6.5
Fine-Grained & Secondary States			
Excited	22.8	**31.7**	+8.9
Amused	20.8	**31.5**	+10.7
Frustration	17.4	**19.8**	+2.4
Annoyance	14.4	**23.8**	+9.4
Depression	**9.0**	1.8	-7.2
Disappointed	10.6	**23.7**	+13.1
Concerned	20.3	**21.0**	+0.7
Confused	33.5	**43.1**	+9.6
Macro F1-Score	*18.6*	***25.1***	*+6.5*

subtle or minority classes. We observe a near-complete collapse in the recognition of "Depressed" (-7.2%) and a significant regression in "Contempt" (-6.5%). This suggests that while the model effectively optimizes for distinct emotional archetypes, it struggles to isolate highly ambiguous, low-arousal, or complex-valence states. For instance, the model likely conflates "Depressed" with "Sad" (which conversely improved by +9.5%) due to their significant spectral and prosodic overlap.

The exact mechanisms driving the degradation in "Contempt" are difficult to isolate purely from aggregate class-wise F1 scores. Consequently, the specific misclassification patterns and inter-class leakages responsible for this decline will be explored in depth during the subsequent confusion matrix analysis.

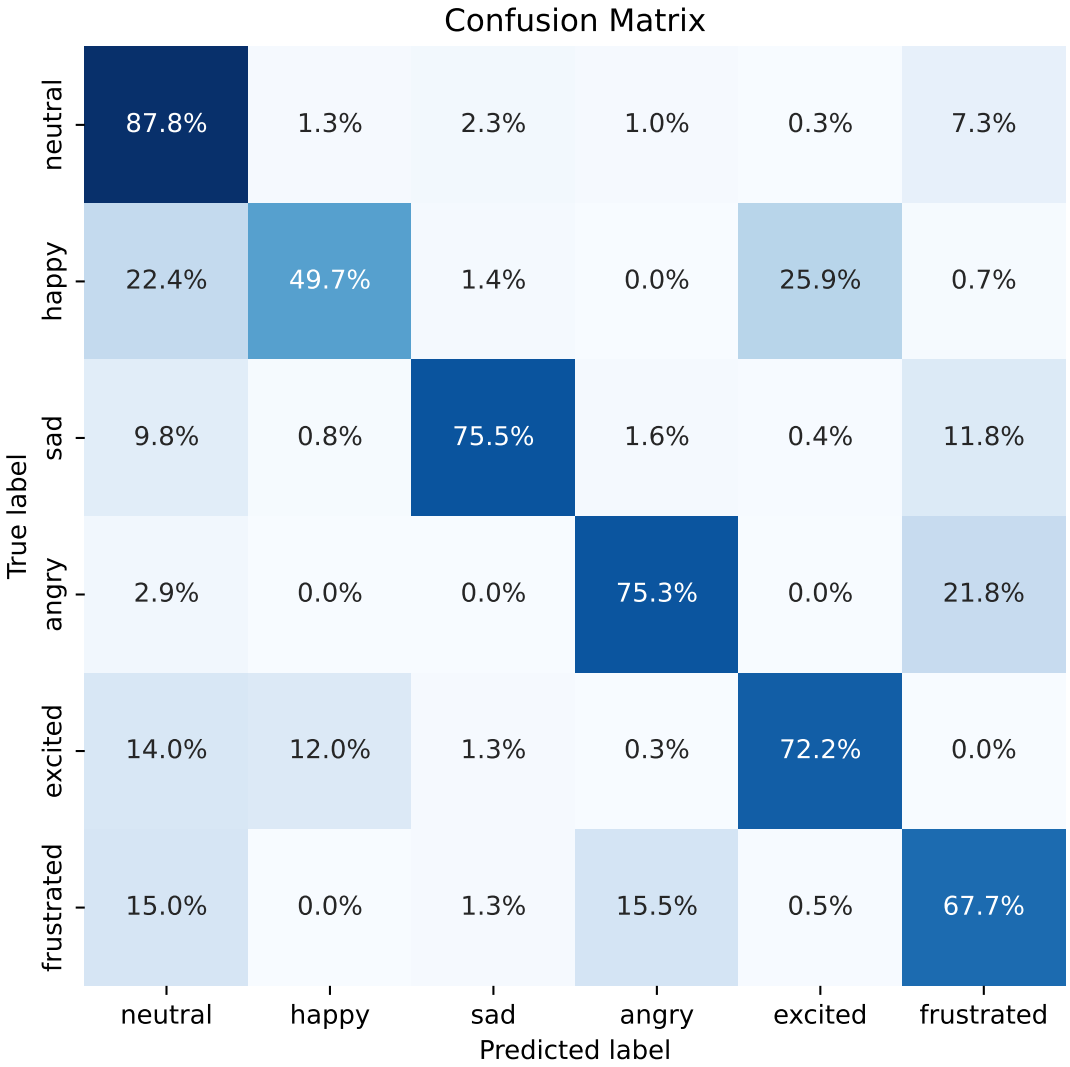

Figure 5.1: Confusion Matrix of Dyadic model for IEMOCAP

5.1.3 Confusion Matrix

IEMOCAP Analysis

To provide a granular view of misclassification patterns, Figure 5.1 presents the confusion matrix for the IEMOCAP dataset. Given the inherent class imbalance (e.g., "Happy" has significantly fewer samples than "Frustrated"), we apply row-wise normalization. This converts raw counts into recall percentages, ensuring that performance on minority classes is explicitly visible and not overshadowed by dominant categories.

Analysis of Misclassification Patterns: The matrix reveals several critical acoustic

behaviors of the model:

- **Distinctiveness of Neutrality:** The model achieves its highest recognition rate on "Neutral" (87.8%). This suggests that the acoustic signature of neutral speech is highly distinct and linearly separable within the high-dimensional feature space, effectively avoiding confusion with active affective states.
- **Positive Valence Ambiguity (Happy ↔ Excited):** A significant bidirectional confusion exists in the positive valence spectrum. Notably, 25.9% of "Happy" samples are misclassified as "Excited", while 12.0% of "Excited" samples leak into "Happy". Furthermore, "Happy" suffers from a severe 22.4% leakage into "Neutral". This dispersion is primarily driven by the low sample count of "Happy" that hinders the model's ability to make accurate predictions.
- **The Anger-Frustration Asymmetry (Impact of Class Imbalance):** A notable directional bias exists in the negative-valence spectrum:
  - **Angry → Frustrated (21.8%):** A substantial portion of "Angry" samples are misclassified as "Frustrated".
  - **Frustrated → Angry (15.5%) and Neutral (15.0%):** Conversely, "Frustrated" samples exhibit a bifurcated leakage, misclassifying almost equally into "Angry" and "Neutral".

  This directional bias is fundamentally driven by the relative distances between these states within the acoustic feature space. "Frustrated" functions as an acoustic "middle ground" situated between "Neutral" and "Angry" on the activation spectrum. This central position explains its near-equal leakage into both the lower-arousal state of "Neutral" (15.0%) and the higher-arousal state of "Angry" (15.5%). Conversely, "Angry" represents an extreme high-activation state. In terms of acoustic identity, it is adjacent to "Frustrated" but significantly distant from "Neutral". Consequently, misclassifications of "Angry" samples leak almost exclusively into the neighboring "Frustrated" category (21.8%), with negligible confusion toward the distant "Neutral" class (2.9%).

MSP-Podcast Analysis

Figure 5.2 presents the row-normalized confusion matrix for the full 16-class taxonomy. Unlike IEMOCAP, where the dataset is heavily skewed towards a few primary emotions, MSP-Podcast presents a more balanced but significantly more complex challenge, requiring the model to discriminate between highly subtle secondary emotional states.

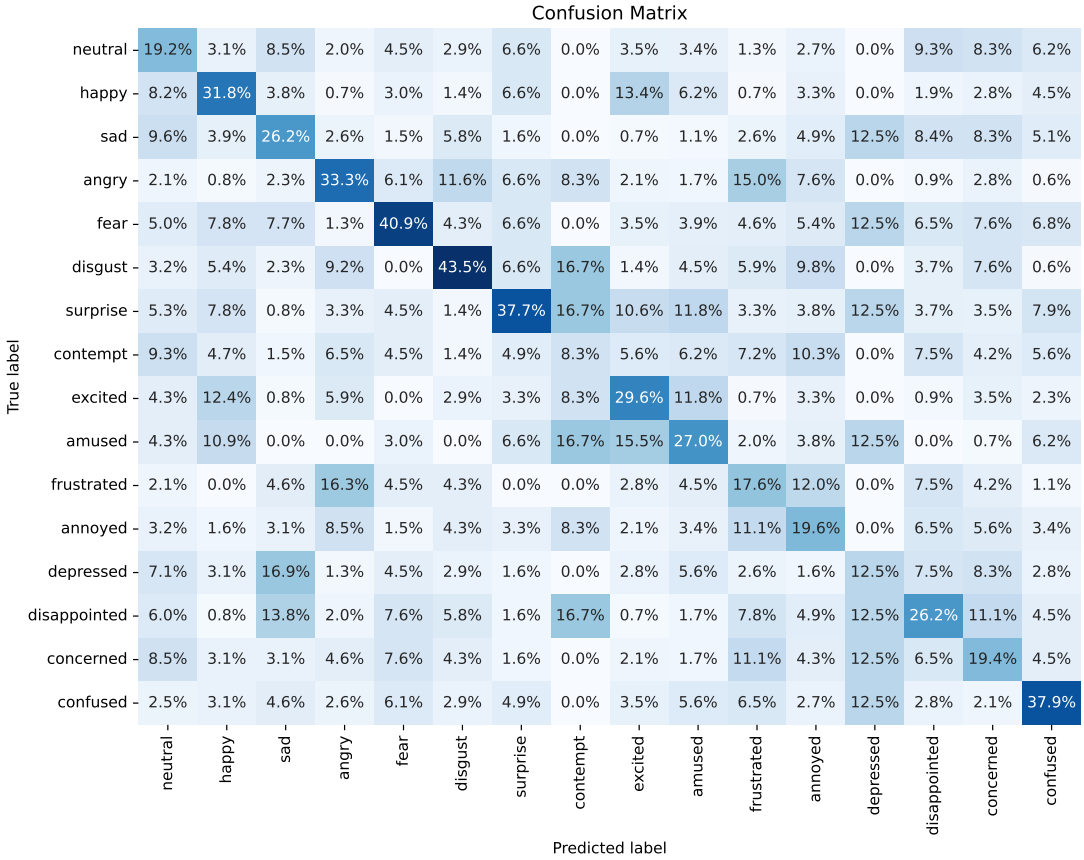

Figure 5.2: Confusion Matrix of Fine-Grained model for MSP-Podcast

Analysis of Misclassification Patterns: The matrix reveals that while the model effectively captures the broad affective dimensions of Valence and Arousal, it encounters specific challenges in disentangling semantically adjacent categories within those quadrants.

- **The "Anger-Frustration-Annoyance" Continuum:** While the framework achieves substantial F1-score improvements in these categories, the confusion matrix reveals that isolating fine-grained variants like "Frustrated" still remains a nuanced

challenge. With a recall of 17.6%, it is frequently misclassified as "Angry" (16.3%) or "Annoyed" (12.0%). This dispersion is not random; all three categories share high-arousal and negative-valence characteristics, forming a distinct "hostility cluster". The model effectively recognizes the vocal agitation and energy but struggles to pinpoint the specific semantic label within this tightly packed acoustic space.

- **Positive Valence Ambiguity:** A similar pattern emerges in the positive spectrum. Despite strong absolute performance gains in "Happy" (+16.6%) and "Amused" (+10.7%), the matrix highlights significant bidirectional leakage among adjacent states. For instance, 12.4% of "Excited" samples are predicted as "Happy", while conversely, 13.4% of "Happy" samples are misclassified as "Excited". This confirms that the model correctly identifies positive activation but often treats these adjacent labels as interchangeable affective synonyms rather than entirely distinct acoustic states.

- **Subtlety of Fine-Grained Emotions:** "Depressed" and "Contempt" prove to be the most elusive categories. "Depressed" samples frequently leak into "Sad" (16.9%), "Concerned" (8.3%), and "Disappointed" (7.5%). This confusion stems directly from acoustic proximity; these emotions all occupy the low-arousal, negative-valence band of the acoustic space. Lacking the distinct high-energy markers of active negative states (like "Anger"), the flat pitch and low intensity of "Depressed" are easily conflated with general sadness or disappointment. Similarly, "Contempt" yields a low recall (8.3%) and does not leak into one specific category, but rather diffuses across the matrix with slight peaks in "Annoyed" (10.3%) and "Neutral" (9.3%). This wide dispersion indicates that "Contempt" is an intrinsically subtle emotion that lacks a unified, highly discriminative acoustic profile, causing the model's predictions to scatter.

Summary: While the granular, on-diagonal accuracy is inherently lower than in the restricted 6-class IEMOCAP task, the off-diagonal errors are heavily concentrated within valid semantic families. The model rarely confuses positive emotions with negative ones (e.g., misclassifications between "Happy" and "Sad" are negligible). This

demonstrates that the learned representations are highly robust in capturing primary affective dimensions (Valence/Arousal), even when the exact fine-grained classification remains ambiguous.

5.2 Cross-Corpus Generalization Analysis

To evaluate the intrinsic robustness of the learned representations, we conduct a Zero-Shot Domain Transfer assessment. In this protocol, the model is trained on one corpus (Source) and evaluated on the other (Target) without any fine-tuning on the target domain.

Table 5.4: Cross-Dataset Evaluation (%). To ensure comparability, the MSP-Podcast test set is filtered to the intersection of classes (Angry, Sad, Happy, Neutral, Excited, Frustrated).

Target Test Set	Training Source	UA ↑	MF1 ↑
MSP-Podcast	IEMOCAP	**38.7**	**35.7**
	MSP-Podcast	32.2	34.9
IEMOCAP	IEMOCAP	**71.4**	**72.0**
	MSP-Podcast	36.2	33.9

Methodology

To ensure a valid comparison despite the divergent emotion taxonomies between the two datasets, we implement specific evaluation protocols tailored to the source model's architecture:

1. Target Label Intersection (Dyadic Model → MSP-Podcast): For the Dyadic model, which is trained on IEMOCAP's restricted 6-class taxonomy, we apply a strict intersection constraint. The model is evaluated exclusively on the subset of MSP-Podcast samples whose ground-truth labels correspond to the six shared emotion classes. This isolates the model's ability to identify "known" categorical emotions within a naturalistic acoustic environment without penalizing it for lacking a vocabulary for additional emotions.

2. Taxonomic Aggregation Protocol (Fine-Grained Model → IEMOCAP): Con-

versely, for the Fine-Grained model trained on MSP-Podcast, we allow the model to predict from its full 16-class taxonomy during inference. However, to achieve a commensurable evaluation against the 6-class IEMOCAP ground-truth, semantically adjacent secondary emotions are aggregated into their primary base categories post-prediction.

This aggregation mapping is firmly grounded in Valence-Arousal (VA) spatial proximity and semantic relatedness:

- **Happy:** Aggregates "Happy" and "Amused" (High Valence, High Arousal).
- **Sad:** Aggregates "Sad", "Depressed", and "Disappointed" (Low Valence, Low Arousal).
- **Angry:** Aggregates "Angry" and "Annoyed" (Low Valence, High Arousal).
- **Direct Mapping:** "Neutral", "Frustrated" and "Excited" map directly to their exact counterparts in the target taxonomy.

Discussion

The results in Table 5.4 reveal a distinct asymmetry in transferability:

1. Robustness of Acted Representations (Dyadic Model → MSP): The most notable finding is that the Dyadic Model outperforms the in-domain Fine-Grained model on the filtered MSP test set (Macro-F1: 35.7% vs. 34.9%). This result serves as strong evidence for the model's generalization capabilities, confirming that it successfully learns robust, context-independent acoustic representations from acted speech.

We attribute this counter-intuitive superiority to the efficiency of prototypical learning. IEMOCAP contains high-intensity, "archetypal" expressions of the six primary emotions, allowing the model to learn sharp decision boundaries. In contrast, the Fine-Grained model is optimized for a diverse 16-class taxonomy. When forced to map its fine-grained internal representations (e.g., "Annoyed", "Concerned") onto a restricted 6-class output space, it likely suffers from increased decision ambiguity compared to the Dyadic model, which was explicitly optimized for this exact taxonomy.

2. The "Context Gap" (Fine-Grained Model → IEMOCAP): Conversely, the Fine-Grained model performs poorly on IEMOCAP (Macro-F1: 33.9%) relative to the in-domain reference (71.4%). We hypothesize that this stems from a structural mismatch in conversation history. MSP-Podcast data often consists of monologues or loose turn-taking, whereas IEMOCAP features rapid, emotionally interdependent exchanges. The Fine-Grained model likely fails to attend to these temporal dependencies, treating the dialogue as isolated segments. This failure underscores the necessity of context-aware training; simply exposing the model to massive "in-the-wild" data is insufficient if that data lacks the dense interactional structure required to learn temporal emotion dynamics.

5.3 Ablation Studies

The results in Table 5.5 reveal critical insights regarding the interplay between prompt structure, feature selection strategies, and VAD dependencies:

Table 5.5: Ablation Studies: Impact of Prompt Structure, Feature Set, and VAD Integration on Model Performance (%). Note: Numeric Selection is utilized for all VAD Integration configurations.

Ablation Setting	**Dyadic Model (IEMOCAP)**		**Fine-Grained Model (MSP-Podcast)**	
	UA ↑	**Macro-F1 ↑**	**UA ↑**	**Macro-F1 ↑**
Prompt Engineering				
Unstructured Prompt	70.63	70.66	18.97	18.57
Structured Prompt	69.96	70.64	19.57	19.10
Feature Selection				
Numeric Selection	**71.36**	**71.99**	**26.55**	**25.13**
Categorical Selection	71.16	71.27	25.86	24.50
LLM-Guided Semantic Selection	70.88	71.08	25.53	24.43
Random Selection	70.48	71.31	25.27	24.33
VAD Integration				
GT VAD	69.37	69.67	23.51	23.47
No VAD	70.87	71.41	24.09	23.06
Stochastic VAD (**Ours**)	**71.36**	**71.99**	**26.55**	**25.13**

1. Effect of Prompt Structure We first evaluate the impact of organizing acoustic

information into a structured Key-Value format versus the baseline's natural language narrative. The empirical results demonstrate divergent effects based on dataset complexity. For the MSP-Podcast task, the structured prompt yields a distinct improvement (e.g., Macro-F1 increases from 18.57% to 19.10%). This suggests that a structured format allows the model's attention mechanism to efficiently isolate key feature variables in complex scenarios. However, on the IEMOCAP dataset, the degradation are negligible (Macro-F1 drops from 70.66% to 70.64%). This indicates that when the prompt contains only a minimal set of acoustic descriptors (the baseline's 5 features) and with the inclusions of historical conversations, a structured format offers limited benefit.

2. Effect of Feature Selection Strategies Next, we analyze the impact of various feature selection methodologies applied to the eGeMAPS. Crucially, the proposed **Numeric Selection** yields the best overall performance, peaking at a Macro-F1 of 71.99% (IEMOCAP) and 25.13% (MSP-Podcast). It consistently outperforms Categorical, LLM-driven, and Random selection methods. This finding demonstrates that numeric feature selection mathematically identifies the optimal feature set for the task, and that while LLMs primarily operate via semantic understanding, they are highly capable of utilizing these quantitative features for effective decision-making.

3. Effect of VAD Integration Finally, we analyze the impact of including continuous VAD scores in the modeling pipeline. The results demonstrate a clear performance drop when the model is trained with GT VAD but forced to rely on predicted VAD during inference (Macro-F1 drops to 69.67% on IEMOCAP and 23.47% on MSP-Podcast). This degradation stems from a critical distribution shift in VAD scores: the model learns to heavily depend on perfect VAD labels as a crutch during training, causing its reasoning to collapse when exposed to the noisy, real-world predictions of the upstream VAD regressor during inference. A deeper exploration of this dependency is provided in Section 5.5.

Conversely, the **Numeric with No VAD** configuration achieves competitive performance while relying solely on the acoustic feature set. While the improvement over the SpeechCueLLM baseline for the Fine-Grained Model is notable (Macro-F1 18.57% →

23.06%), the effect on the Dyadic Model is marginal (Macro-F1 70.66% → 71.41%). This indicates that while the selected acoustic features are strong, they alone cannot fully resolve complex emotional ambiguities.

Crucially, our proposed **Numeric with Stochastic VAD** strategy successfully overcomes these limitations, achieving the highest overall performance across both datasets (peaking at a Macro-F1 of 71.99% on IEMOCAP and 25.13% on MSP-Podcast). By randomly substituting GT VAD with predicted scores during training and injecting discrete noise, the model learns to treat the dimensional coordinates as soft, guiding heuristics rather than absolute rules. This regularization effectively bridges the gap between the clean training environment and noisy inference conditions, allowing the LLM to leverage the highly discriminative power of VAD without falling into the dependency trap.

5.4 VAD Upper Bound Analysis

To establish the theoretical upper bound of the proposed framework, we conduct an Upper Bound Analysis by evaluating the model's performance with GT VAD scores during inference. We then contrast this ideal scenario against the model's performance when relying on predicted VAD scores. This comparison is critical because, in real-world deployment, obtaining perfectly accurate dimensional annotations is highly improbable, inherently exposing the system to downstream error propagation.

5.4.1 VAD Regression Performance (CCC Analysis)

Before evaluating the downstream classification performance using predicted VAD scores, it is crucial to assess the reliability of the upstream regressor itself.

Table 5.6: Concordance Correlation Coefficient (CCC) of the VAD Regressor. Lower CCC scores indicate higher noise in the predicted primitives.

Dataset	**Valence** ↑	**Arousal** ↑	**Dominance** ↑
IEMOCAP	0.7162	0.7876	0.6258
MSP-Podcast	0.6214	0.6752	0.5749

As evident in Table 5.6, the regression performance exhibits a distinct disparity across the affective dimensions:

- **Arousal is robust:** Arousal achieves the highest CCC across both datasets (0.7876 and 0.6752, respectively). This is likely because acoustic energy and pitch variation serve as strong, linear indicators of activation, which the regression model captures effectively.
- **Dominance is the bottleneck:** Conversely, the CCC for Dominance is significantly lower, bottoming out at 0.5749 on the naturalistic MSP-Podcast dataset. The regressor struggles to distinguish strong from weak dominance based solely on acoustics, highlighting the difficulty of modeling complex interpersonal dynamics without semantic or visual context.

This variance in regression reliability—particularly the noise present in the Valence and Dominance predictions—directly foreshadows the distribution shift challenges the LLM encounters during the downstream classification task.

5.4.2 Result Analysis

Table 5.7 presents the downstream classification results of the evaluation. When provided with perfect dimensional information (ground-truth VAD) during both training and inference, the framework achieves its theoretical upper bound, with Macro-F1 surging to 73.90% on IEMOCAP and 31.17% on MSP-Podcast.

Table 5.7: Upper Bound Analysis: Impact of substituting GT VAD with predicted VAD during inference. Both configurations use the Numeric feature set and are trained exclusively on GT VAD.

	IEMOCAP		**MSP-Podcast**	
Inference Condition (Trained on GT)	**UA ↑**	**MF1 ↑**	**UA ↑**	**MF1 ↑**
Evaluate with GT VAD (Upper Bound)	**74.17**	**73.90**	**31.46**	**31.17**
Evaluate with Pred VAD	65.80	66.40	23.51	23.47

These results unequivocally confirm that VAD scores are highly potent indicators of discrete emotional states. However, this configuration relies on an unrealistic assumption: the availability of perfect affective labels during real-world inference.

When the evaluation shifts to using predicted VAD scores, we observe a substantial performance degradation. On the IEMOCAP dataset, Macro-F1 drops sharply to 66.40%. This phenomenon is an example of error propagation and distribution shift: the LLM learns to over-rely on the perfect VAD scores as heuristic "shortcuts" during training. During inference, these shortcuts become noisy or inaccurate due to the upstream regressor, hence the model's reasoning collapses.

A similar degradation occurs on the MSP-Podcast dataset, where performance drops relative to the setting (Macro-F1 31.17% → 23.47%). However, it is noteworthy that even this degraded model still outperforms the SpeechCueLLM baseline (Macro-F1 18.57%). This suggests that for naturalistic data, even noisy dimensional priors provide valuable directional cues that help disentangle complex emotional states.

Conversely, in the cleaner, acted IEMOCAP environment, the introduction of predictive noise outweighs the informational gain.

5.5 Impact of ASR Errors

To evaluate the model's resilience to transcription errors, we replace the ground truth text for both the historical conversation and the target utterance with hypotheses generated by *Whisper-v3-large-turbo*.

Crucially, we utilize the raw ASR output without standard text normalization (e.g., stripping punctuation or lowercasing). We hypothesize that orthographic features such as exclamation points, ellipses, and capitalization often encode vital paralinguistic cues that contribute to emotional interpretation.

Dataset	**Raw Text WER ↓**	**Normalized Text WER ↓**
IEMOCAP	32.27	15.48
MSP-Podcast	34.01	14.06

Table 5.8: Word Error Rate (WER) Analysis of Whisper-v3-large-turbo. (%)

WER Analysis

Table 5.8 presents the error rates for the ASR model. We observe a significant disparity between the Raw WER (> 32%) and the Normalized WER (< 15.5%). This delta indicates that approximately 50% of the measured "errors" stem from stylistic divergences—such as punctuation placement or capitalization styles—rather than semantic corruptions. Consequently, the core linguistic content required for emotion inference remains largely intact.

5.5.1 Performance Robustness

To quantify the downstream impact of these ASR errors, we compare the model's performance when conditioned on GT versus ASR hypotheses.

		IEMOCAP		MSP-Podcast	
Model	**Text Source**	**UA ↑**	**MF1 ↑**	**UA ↑**	**MF1 ↑**
Proposed Framework	Ground Truth	**71.36**	**71.99**	**26.55**	**25.13**
	ASR Hypothesis	69.67	70.54	25.89	24.41

Table 5.9: Impact of ASR Error on Model Performance. We compare the robustness of the proposed model when using GT transcripts versus ASR hypotheses.

As demonstrated in Table 5.9, the impact of ASR noise varies across domains. On the IEMOCAP dataset, we observe an expected degradation pattern, with the Macro-F1 declining by 1.45% absolute (71.99% → 70.54%). This drop is consistent with the introduction of transcription errors into a controlled environment where the ground-truth text is highly accurate and grammatically intact.

Conversely, the Macro-F1 for the Fine-Grained model on the MSP-Podcast dataset degrades by only 0.72% (25.13% → 24.41%), demonstrating significantly greater resilience than the Dyadic model. We hypothesize that this stems from the specific segmentation characteristics of the MSP-Podcast corpus, where audio clipping often results in truncated or grammatically incomplete sentences in the ground-truth. Generative ASR models, in their attempt to transcribe, frequently perform implicit "semantic completion." By effectively guessing the missing context to produce syntactically coherent sentences, the ASR inadvertently provides the LLM with a more

structured linguistic input, thereby buffering the overall detrimental effect of the WER on downstream emotion recognition.

5.6 Impact of Gender Misclassifications

Because our feature extraction pipeline relies on gender-based standardization for the numeric acoustic features, the downstream SER performance is inherently coupled with the accuracy of the upstream gender classifier. To evaluate the framework's resilience to gender misclassification in a fully automated pipeline, we replace GT gender labels with predictions generated by our fine-tuned W2V2 classifier.

Dataset	**WA** $\uparrow$
IEMOCAP	99.0
MSP-Podcast	90.0

Table 5.10: Performance of W2V2 Gender Classifier. (%)

Model Accuracy and Fallback Strategy

Table 5.10 details the upstream performance of the gender classifier. On the dyadic IEMOCAP dataset, the model achieves near-perfect accuracy (99.0%), allowing us to confidently utilize its discrete predictions directly for feature standardization. However, performance degrades to 90.0% on the MSP-Podcast dataset, a drop attributed to the extreme acoustic diversity and the presence of hundreds of unconstrained speakers in naturalistic environments.

To mitigate the risk of applying incorrect gender normalizations—which would heavily distort the continuous acoustic features before they reach the LLM—we introduce a probabilistic fallback mechanism for the naturalistic dataset. Rather than forcing a hard binary prediction, we evaluate the classifier's output probability, p. If the prediction falls within an uncertainty boundary of $0.2 \leq p \leq 0.8$, the system abstains from making a definitive gender assignment. In these ambiguous cases, the pipeline defaults to a general, gender-agnostic standardization computed across the entire combined training corpus. This conservative thresholding approach ensures that acoustically ambiguous

voices are not penalized by erroneously biased normalization parameters.

5.6.1 Performance Robustness

To evaluate the robustness of the proposed framework in a fully automated pipeline, we compare the downstream emotion classification performance when the acoustic features are standardized using GT gender labels versus predicted gender.

		IEMOCAP		MSP-Podcast	
Model	**Experiment**	**UA ↑**	**MF1 ↑**	**UA ↑**	**MF1 ↑**
Proposed Framework	Ground Truth	**71.36**	**71.99**	**26.55**	**25.13**
	Pred Gender	70.96	71.73	26.49	25.09

Table 5.11: Impact of predicted gender standardization on downstream SER performance. (%)

As demonstrated in Table 5.11, the framework exhibits strong resilience to upstream gender classification errors, though the impact varies across datasets. On the dyadic IEMOCAP dataset, despite the gender classifier achieving near-perfect accuracy (99.0%), we observe a marginal degradation in Macro-F1 of 0.26 percentage points (71.99% → 71.73%). This slight drop indicates that even rare misclassifications in an otherwise pristine dataset can introduce minor distributional shifts in the standardized features.

Conversely, for the Fine-Grained Model (MSP-Podcast), the Macro-F1 degrades by a mere 0.04 percentage points (25.13% → 25.09%). Despite the upstream gender classifier operating at a lower accuracy of 90.0%, our proposed probabilistic fallback strategy minimized the impact. By dynamically applying gender-agnostic standardization to acoustically ambiguous speakers, the hybrid framework successfully neutralizes potential error propagation, proving its robustness and adaptability in noisy, naturalistic scenarios.

5.7 Qualitative Examples

To illustrate the practical impact of the proposed feature integration, we present a case study comparing the decision-making process of the baseline model versus the

proposed framework on a challenging sample from both test sets.

5.7.1 MSP-Podcast Prompt

Context: The target utterance involves a speaker expressing frustration about a legal injustice. The text itself is semantically ambiguous, plausibly supporting an interpretation of either "Anger" or "Disappointed".

Baseline Prediction (Failure Case)

Baseline Prompt

Prompt Input: Now you are expert of sentiment and emotional analysis. The following noted between '### ###' is a single isolated utterance with its speech features attached. ###
Speaker_0:"yeah, it's not fair. i think the saddest thing is that there's law that lets them do this. there's literal law that lets them do this." (medium pitch with high variation) ###
Target speech characteristics: moderate volume with moderate variation, moderate pitch with high variation, moderate speaking rate.
Please select the emotional label of "yeah, it's not fair. i think the saddest thing is that there's law that lets them do this. there's literal law that lets them do this." from [angry, frustrated, disgust, annoyed, sad, depressed, disappointed, fear, happy, surprise, excited, contempt, amused, concerned, confused, neutral] based on both the context and audio features. Respond with just one label:

Model Output: `anger`

Ground Truth Label: `disappointed`

Analysis: The baseline model defaults to "Angry". This is likely driven by the semantic content ("it's not fair"), which strongly implies injustice—a common trigger for anger. The vague acoustic descriptor ("moderate pitch with high variation") is insufficient to contradict this textual bias.

Proposed Framework (Success Case)

Proposed Framework Prompt (Condensed)

Prompt Input: *[...Standard Instructions Omitted...]*

Target speech characteristics:

Valence: 2.6/7 Arousal: 4.8/7 Dominance: 4.6/7

... *[Pitch features omitted]* ...

— Loudness —

Loudness Variation (StdDev): **Low**

... *[Other Loudness features omitted]* ...

— Voice Quality —

Jitter (Voice Roughness): **High**

Shimmer (Voice Breathiness): Medium

Transcript: "yeah, it's not fair. i think the saddest thing..."

Please select the emotional label... *[...remainder of prompt identical...]*

Model Output: `disappointed`

Ground Truth Label: `disappointed`

Analysis: The proposed model correctly predicts "Disappointed". This correction is driven by specific feature cue:

- **Moderate Arousal (4.8/7):** While "Anger" is typically associated with extreme arousal (high energy), this moderate score lowers the probability of a high-activation prediction, steering the model toward resignation rather than aggression.
- **High Jitter:** The inclusion of voice quality metrics is decisive here. High jitter indicates vocal instability or roughness, a prosodic feature frequently correlated with sadness, grief, and disappointment.

5.7.2 IEMOCAP Prompt

Context: The target conversation features two speakers stranded without essential supplies, such as a flashlight or warm clothing. The lexical content is highly ambiguous: the exchange can be interpreted either as hostile sarcasm driven by acute frustration, or as playful, morale-boosting banter used to cope with an uncomfortable situation.

Baseline Prediction (Failure Case)

Baseline Prompt

Prompt Input: Now you are expert of sentiment and emotional analysis. The following conversation noted between '### ###' involves several speakers. The last three utterances are followed by its speech features. ###
Speaker_1:"What flashlight?"
Speaker_0:"The flashlight, the silver one, there's only one, isn't there?"
Speaker_1:"It's not your flashlight."
Speaker_0:"Whose is it?"
Speaker_0:"Well, Sure."
Speaker_1:"I mean you keep saying,my flashlight, as if it's somehow just yours. How's that suppose to make me feel?" (medium pitch with medium variation)
Speaker_0:"No, Of course, you are right. Our flashlight. Me flashlight is you flashlight naturally. How can we not think to bring it?" (medium pitch with medium variation)
Speaker_1:"yes." (very_low pitch with medium variation)
Speaker_0:"Are you cold? You want my jacket? We should have bought a blanket. Our blanket-. This is great, isn't it?" (medium pitch with medium variation) ###

Target speech characteristics: moderate volume with high variation, moderate pitch with moderate variation, very high speaking rate.
Please select the emotional label of [Speaker_0:"Are you cold? You want my jacket? We should have bought a blanket. Our blanket-. This is great, isn't it?"]

from [happy, sad, neutral, angry, excited, frustrated] based on both the context and audio features. Respond with just one label:

Model Output: `neutral`

Ground Truth Label: `excited`

Analysis: The baseline model defaults to a "Neutral" classification. Due to the profound semantic ambiguity of the conversational history, the model is unable to confidently derive an affective trajectory, forcing it to rely heavily on the target utterance. Lexically, the target utterance suggests the speaker is utilizing sarcastic banter to cope with an uncomfortable environment. However, without explicit emotional trigger words, the LLM exhibits lexical bias and maps this coping mechanism to a baseline, unemotional state. Furthermore, the limited macro-acoustic descriptors provided to the baseline—where only a very high speaking rate" stands out—result in feature starvation. This sparse acoustic evidence is fundamentally insufficient to override the model's text-based assumption, compelling it to default to a safe, "Neutral" conclusion.

Proposed Framework (Success Case)

Proposed Framework Prompt (Condensed)

Prompt Input: *[...Standard Instructions Omitted...]*

Target speech characteristics:

Valence: 4.5/5 Arousal: 4.0/5 Dominance: 3.5/5

... *[Pitch features omitted]* ...

... *[Loudness features omitted]* ...

— Voice Quality (Roughness/Breathiness) —

Jitter (Voice Roughness): High

Shimmer (Voice Breathiness): High

— MFCCs (Abstract Spectral Shape) —

MFCC 1 (Overall Spectral Shape): Medium

MFCC 1 (Voiced Spectral Shape): Medium

MFCC 1 Variation (Overall): High

MFCC 1 Variation (Voiced): High

Transcript: "Are you cold? You want my jacket? We should have bought a blanket. Our blanket-. This is great, isn't it?"

Please select the emotional label... *[...remainder of prompt identical...]*

Model Output: `excited`

Ground Truth Label: `excited`

Analysis: The proposed model correctly predicts "Excited," demonstrating how the integration of micro-level features fundamentally alters the affective interpretation. This accurate classification is driven by specific quantitative cues:

- **High Positive Valence (4.5/5):** This dimensional metric serves as a definitive anchor. It immediately resolves the profound semantic ambiguity of the conversational history, overriding the baseline's "Neutral" assumption and confirming a highly positive affective state.
- **High Jitter, Shimmer, and MFCC 1 Variation:** While the high valence narrows the prediction to either "Happy" or "Excited," the inclusion of these specific voice quality and spectral metrics provides the decisive signal. High Jitter and Shimmer capture the acoustic turbulence of breathless, giddy laughter, while high MFCC 1 variation indicates a highly animated vocal tract. This energetic acoustic signature steers the model toward "Excited" rather than a calmer "Happy."

5.7.3 Summary

These comparisons illustrate a fundamental advantage of the proposed framework. While the baseline's reliance on broad qualitative descriptors (e.g., "medium pitch", "high variation") captures general acoustic trends, it often lacks the resolution to distinguish between emotionally adjacent states.

By integrating precise, quantitative feature values—such as continuous dimensional

predictions and micro-level acoustic metrics—the proposed approach provides the necessary discriminative signal to resolve these ambiguities. Consequently, this enables the model to look beyond the semantic content of the transcript and accurately identify nuanced emotional expressions that generic descriptors fail to capture.

Chapter 6

Conclusion & Future Work

This chapter summarizes the core contributions and empirical findings of the research before outlining potential avenues for future exploration.

6.1 Conclusion

This research presented a novel, automated Multimodal SER framework that shifts SER from an opaque "black box" into a transparent, interpretable reasoning pipeline. By integrating the generative reasoning capabilities of LLMs with a structured, quantitative acoustic representation, the proposed system systematically addresses the critical limitations of existing baselines—specifically their reliance on coarse-grained taxonomies, feature scarcity, and inefficient prompting mechanisms. To resolve the "semantic bottleneck" where acoustically similar emotions become indistinguishable, we introduced a hybrid feature selection strategy incorporating up to 30 most discriminative descriptors from eGeMAPS. Simultaneously, we validated that transitioning from unstructured narratives to a Structured Key-Value Prompting strategy significantly mitigates information dilution, enhancing the model's ability to attend to these high-dimensional paralinguistic cues.

Experimental validation confirmed the efficacy of this approach across disparate domains. On the IEMOCAP dataset, the model achieved an **Macro-F1 of 71.99%**, representing a **1.33% improvement** over the SpeechCueLLM baseline. On the more

challenging, naturalistic MSP-Podcast corpus, the framework reached a **Macro-F1 score of 25.13%**, achieving a substantial **6.56% improvement** and demonstrating exceptional capability in recognizing fine-grained secondary emotions. Furthermore, zero-shot cross-corpus evaluations confirmed that the Dyadic Model effectively generalizes learned representations from acted training data to naturalistic test environments.

Finally, ablation studies and error propagation analysis validated the critical importance of Stochastic VAD Substitution, proving that the framework is highly resilient to downstream error propagation from upstream ASR and gender classification modules. Ultimately, this work establishes that augmenting foundation models with precise, structured paralinguistic features provides a highly scalable and domain-agnostic solution for robust emotion recognition. By ensuring these affective predictions remain inherently interpretable, this framework paves the way for more intuitive, and justifiable Human-Computer Interaction (HCI) systems.

6.2 Future Work

While the proposed framework establishes a strong baseline for LLM-based SER, several avenues for future research remain to further enhance performance and scalability.

6.2.1 Architectural Scaling with Next-Generation LLMs

The current implementation relies on **Llama-3-8B-Instruct**, which offers a balance of efficiency and performance. However, recent advancements in foundation models suggest that newer architectures (e.g., Llama-3-70B, GPT-4o) possess better reasoning capabilities. Future work should investigate whether replacing with newer models unlocks emergent abilities, such as the capacity to disambiguate complex mixed-emotional states without extensive fine-tuning.

6.2.2 Probabilistic Multi-Label Prediction

Human emotion is rarely monolithic; natural speech often conveys a blend of overlapping affective states (e.g., Surprised yet Happy). The current framework frames

SER as a single-label classification task, forcing the model to select one exclusive category. However, the MSP-Podcast dataset provides soft probability distributions across multiple emotions. Future iterations of the model could be trained to output a ranked dictionary of the top-3 probable emotions rather than a single tag. This multi-label approach would better reflect the ambiguity of human affect and provide downstream applications with a more nuanced understanding of the user's emotional state.

6.2.3 Weakly Supervised Learning with MSP-Conversation

Data scarcity remains a significant bottleneck for training robust SER models, particularly for naturalistic speech. The **MSP-Conversation** corpus [62] offers a vast repository of "in-the-wild" recordings that complements the MSP-Podcast dataset. Unlike the isolated, sentence-level segments of MSP-Podcast which often miss the temporal context of dialogue, MSP-Conversation consists of long-form interactions (10–20 minutes) annotated with time-continuous emotional traces for Valence (Negative to Positive), Arousal (Calm to Active), and Dominance (Weak to Strong).

A critical feature of this corpus is that its recordings are sourced from the same podcast pool as the MSP-Podcast database. This acoustic overlap is particularly advantageous for our proposed future work, as it ensures that representations learned from MSP-Conversation are directly transferable to MSP-Podcast without introducing domain shift artifacts. Furthermore, the corpus construction methodology explicitly balances emotional and demographic representation by selectively annotating segments known to contain diverse affective content.

To leverage this unutilized resource, future work could implement a Weak Supervision strategy. Since the corpus is primarily annotated with dimensional scores rather than categorical labels, we propose developing a mapping function (either rule-based or clustering-based) to convert continuous VAD scores into pseudo-categorical emotion tags. This approach would allow us to combine MSP-Conversation, IEMOCAP, and MSP-Podcast into a massive, unified training dataset. Such a large-scale pre-training regime has the potential to significantly improve the model's generalization capabilities across diverse acoustic environments while incorporating the long-term temporal

dependencies essential for realistic emotion recognition.

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Appendix A

Curated Acoustic Feature Sets (eGeMAPS)

This appendix details the comprehensive acoustic feature sets obtained during the preprocessing phase. Because the proposed framework dynamically adapts to different acoustic environments, the specific subset of eGeMAPS features curated for the IEMOCAP dataset differs from the subset curated for the MSP-Podcast dataset.

To establish the interpretable reasoning pathway central to the framework, the raw feature names generated by the openSMILE toolkit are mapped to human-readable names.

Note: The definitions and functional descriptions in this dictionary are strictly adapted from the original eGeMAPS specification and official openSMILE documentation.

A.1 IEMOCAP Feature Set

The following features were selected via the hybrid Lasso Regression and ANOVA mechanism as the most discriminative paralinguistic markers for the IEMOCAP dataset.

Pitch (Intonation)

- **`F0semitoneFrom27.5Hz_sma3nz_amean`** → **Average Pitch**:The arithmetic mean of the fundamental frequency across voiced speech segments. To align with human auditory perception, F0 is extracted on a logarithmic semitone scale (relative to 27.5 Hz) and temporally smoothed using a 3-frame moving average filter.
- **`F0semitoneFrom27.5Hz_sma3nz_stddevNorm`** → **Pitch Stability (StdDev)**: The standard deviation of the fundamental frequency, normalized by its arithmetic mean (coefficient of variation).
- **`F0semitoneFrom27.5Hz_sma3nz_pctlrange0-2`** → **Pitch Range (Dynamic Range)**: The strict distance between the 80th and 20th percentiles of the semitone-scaled fundamental frequency.

Loudness (Energy)

- **`loudness_sma3_amean`** → **Average Loudness**: The arithmetic mean of the perceived speech intensity derived from the auditory spectrum, computed across both voiced and unvoiced regions to capture the total utterance energy.
- **`equivalentSoundLevel_dBp`** → **Overall Sound Level (dBP)**: The continuous equivalent sound pressure level (LEq) of the utterance, expressed in decibels (dB).
- **`loudness_sma3_stddevNorm`** → **Loudness Variation (StdDev)**: The standard deviation of the perceived speech intensity, normalized by its arithmetic mean.
- **`loudness_sma3_meanRisingSlope`** → **Avg. Loudness Increase Slope**: The arithmetic mean of the positive first derivative (rising slopes) of the perceived speech intensity.
- **`loudness_sma3_stddevFallingSlope`** → **Loudness Decrease Variability**: The standard deviation of the negative first derivative (falling slopes) of the perceived speech intensity.

Rhythm & Speed

- `VoicedSegmentsPerSec` → **Speaking Rate**: The number of continuously voiced regions per second, serves as a pseudo-syllable rate, directly correlates with the perceived tempo of speech.
- `StddevUnvoicedSegmentLength` → **Unvoiced Length Variation**: The standard deviation of the duration of continuously unvoiced regions ($F0 = 0$).

Voice Quality (Roughness & Breathiness)

- `jitterLocal_sma3nz_amean` → **Jitter (Voice Roughness)**: The arithmetic mean of the local jitter, defined as the deviations in individual consecutive F0 period lengths.
- `shimmerLocaldB_sma3nz_amean` → **Shimmer (Voice Breathiness)**: The arithmetic mean of the local shimmer, defined as the difference in the peak amplitudes of consecutive F0 periods.
- `HNRdBACF_sma3nz_amean` → **Harmonics-to-Noise Ratio**: The arithmetic mean of the Harmonics-to-Noise Ratio, quantifies the relation of acoustic energy in harmonic components to the energy in noise-like components.
- `hammarbergIndexV_sma3nz_stddevNorm` → **Hammarberg Variation**: Standard deviation of the Hammarberg Index, normalized by its arithmetic mean, measures the variability of the ratio between the strongest acoustic energy peak in the 0–2 kHz region and the strongest peak in the 2–5 kHz region.

Spectral Balance (Timbre & Tone)

- `alphaRatioV_sma3nz_amean` → **Alpha Ratio (Voiced)**: The arithmetic mean of the Alpha Ratio. Computed exclusively across voiced regions, quantifies the ratio of the summed acoustic energy in the 50–1000 Hz frequency band to the summed energy in the 1–5 kHz band.
- `alphaRatioUV_sma3nz_amean` → **Alpha Ratio (Unvoiced)**: The arithmetic

mean of the Alpha Ratio. Computed exclusively across unvoiced regions, quantifies the ratio of the summed acoustic energy in the 50–1000 Hz frequency band to the summed energy in the 1–5 kHz band.

- **`slopeV0-500_sma3nz_amean` → Low Freq Spectral Slope (Voiced)**:The arithmetic mean of the low-frequency spectral slope. Computed exclusively across voiced regions, calculates the linear regression slope of the logarithmic power spectrum within the 0–500 Hz frequency band.
- **`slopeUV0-500_sma3nz_amean` → Low Freq Spectral Slope (Unvoiced)**: The arithmetic mean of the low-frequency spectral slope. Computed exclusively across unvoiced regions, calculates the linear regression slope of the logarithmic power spectrum within the 0–500 Hz frequency band.
- **`spectralFlux_sma3_stddevNorm` → Overall Spectral Flux Variation**: The standard deviation of the spectral flux, normalized by its arithmetic mean. Computed across both voiced and unvoiced regions, quantifies the variation in the difference of the spectra between two consecutive frames.
- **`spectralFluxV_sma3nz_stddevNorm` → Spectral Flux Variation (Voiced)**: The standard deviation of the spectral flux, normalized by its arithmetic mean. Computed exclusively across voiced regions, quantifies the variation in the difference of the spectra between two consecutive frames.

Formants (Vowel Articulation)

- **`F1frequency_sma3nz_amean` → F1 Frequency (Vowel Openness)**: The arithmetic mean of the first formant.
- **`F1amplitudeLogRelF0_sma3nz_amean` → F1 Amplitude**: The arithmetic mean of the energy ratio between the spectral harmonic peak at the first formant's center frequency and the spectral peak at F0.

MFCCs (Abstract Spectral Shape)

- **`mfcc1_sma3_amean` → MFCC 1 (Overall Spectral Shape)**: The arithmetic mean of MFCC 1.
- **`mfcc1V_sma3nz_amean` → MFCC 1 (Voiced Spectral Shape)**: The arithmetic mean of the MFCC 1, computed exclusively across voiced regions.
- **`mfcc1_sma3_stddevNorm` → MFCC 1 Variation (Overall)**: The standard deviation of MFCC 1, normalized by its arithmetic mean.
- **`mfcc1V_sma3nz_stddevNorm` → MFCC 1 Variation (Voiced)**: The standard deviation of MFCC 1, normalized by its arithmetic mean, computed exclusively across voiced regions.
- **`mfcc2V_sma3nz_amean` → MFCC 2 (Voiced Spectral Energy)**: The arithmetic mean of the MFCC 2, computed exclusively across voiced regions.
- **`mfcc4_sma3_amean` → MFCC 4 (Overall High-Freq Detail)**: The arithmetic mean of the MFCC 4.
- **`mfcc4V_sma3nz_amean` → MFCC 4 (Voiced High-Freq Detail)**: The arithmetic mean of the MFCC 4, computed exclusively across voiced regions.

A.2 MSP-Podcast Feature Set

The following features were curated specifically to capture the highly nuanced, spontaneous emotional expressions characteristic of the naturalistic MSP-Podcast dataset.

Pitch (Intonation)

- **`F0semitoneFrom27.5Hz_sma3nz_amean` → Average Pitch**: The arithmetic mean of the fundamental frequency across voiced speech segments. To align with human auditory perception, F0 is extracted on a logarithmic semitone scale

(relative to 27.5 Hz) and temporally smoothed using a 3-frame moving average filter.

- **`F0semitoneFrom27.5Hz_sma3nz_stddevNorm` → Pitch Stability (StdDev)**: The standard deviation of the fundamental frequency, normalized by its arithmetic mean (coefficient of variation).

Loudness (Energy & Dynamics)

- **`loudness_sma3_amean` → Average Loudness**: The arithmetic mean of the perceived speech intensity derived from the auditory spectrum, computed across both voiced and unvoiced regions to capture the total utterance energy.
- **`equivalentSoundLevel_dBp` → Overall Sound Level (dBP)**: The continuous equivalent sound pressure level (LEq) of the utterance, expressed in decibels (dB).
- **`loudness_sma3_pctlrange0-2` → Loudness Range**: The strict distance between the 80th and 20th percentiles of the perceived speech intensity, derived from the auditory spectrum and computed across both voiced and unvoiced regions.
- **`loudness_sma3_stddevNorm` → Loudness Variation (StdDev)**: The standard deviation of the perceived speech intensity, normalized by its arithmetic mean.
- **`loudness_sma3_meanRisingSlope` → Avg. Loudness Increase Slope**: The arithmetic mean of the positive first derivative (rising slopes) of the perceived speech intensity.
- **`loudness_sma3_meanFallingSlope` → Avg. Loudness Decrease Slope**: The arithmetic mean of the negative first derivative (falling slopes) of the perceived speech intensity.
- **`loudness_sma3_stddevFallingSlope` → Loudness Decrease Variability**: The standard deviation of the negative first derivative (falling slopes) of the perceived speech intensity.

- **loudnessPeaksPerSec → Loudness Peaks per Second**: The number of loudness peaks per second.
- **loudness_sma3_percentile20.0 → Loudness 20th Percentile**: The 20th percentile of the perceived speech intensity.

Spectral Balance (Timbre & Tone)

- **alphaRatioV_sma3nz_amean → Alpha Ratio (Spectral Balance)**: The arithmetic mean of the Alpha Ratio. Computed exclusively across voiced regions, quantifies the ratio of the summed acoustic energy in the 50–1000 Hz frequency band to the summed energy in the 1–5 kHz band.
- **slopeV500-1500_sma3nz_amean → Spectral Slope (500-1500 Hz)**: The arithmetic mean of the mid-frequency spectral slope. Computed exclusively across voiced regions, calculates the linear regression slope of the logarithmic power spectrum within the 500–1500 Hz frequency band.
- **spectralFluxV_sma3nz_amean → Spectral Flux (Timbre Change)**: The arithmetic mean of the spectral flux. Computed exclusively across voiced regions, quantifies the variation in the difference of the spectra between two consecutive frames.
- **spectralFluxUV_sma3nz_amean → Spectral Flux (Unvoiced Regions)**: The arithmetic mean of the spectral flux. Computed exclusively across unvoiced regions, quantifies the variation in the difference of the spectra between two consecutive frames.
- **spectralFluxV_sma3nz_stddevNorm → Spectral Flux Variation (Voiced)**: The standard deviation of the spectral flux, normalized by its arithmetic mean. Computed exclusively across voiced regions, quantifies the variation in the difference of the spectra between two consecutive frames.
- **hammarbergIndexV_sma3nz_amean → Hammarberg Index (Voice Sharpness)**: The arithmetic mean of the Hammarberg Index. Computed exclusively across voiced regions, quantifies the ratio of the strongest acoustic energy peak

in the 0–2 kHz frequency region to the strongest energy peak in the 2–5 kHz frequency region.

Pace & Timing (Conversational Rhythm)

- `VoicedSegmentsPerSec` → **Speaking Rate**: The number of continuously voiced regions per second, serves as a pseudo-syllable rate, directly correlates with the perceived tempo of speech.
- `MeanUnvoicedSegmentLength` → **Avg. Unvoiced Length**: The arithmetic mean of the duration of continuously unvoiced regions ($F0 = 0$).
- `StddevUnvoicedSegmentLength` → **Unvoiced Length Variation**: The standard deviation of the duration of continuously unvoiced regions ($F0 = 0$).
- `StddevVoicedSegmentLengthSec` → **Voiced Length Variation (StdDev)**: The standard deviation of the duration of continuously voiced regions ($F0 > 0$).

Voice Quality & Formants

- `mfcc1V_sma3nz_amean` → **MFCC 1 (Spectral Shape)**: The arithmetic mean of the MFCC 1, computed exclusively across voiced regions.
- `jitterLocal_sma3nz_amean` → **Jitter (Voice Roughness)**: The arithmetic mean of the local jitter, defined as the deviations in individual consecutive F0 period lengths.
- `shimmerLocaldB_sma3nz_amean` → **Shimmer (Voice Breathiness)**: The arithmetic mean of the local shimmer, defined as the difference in the peak amplitudes of consecutive F0 periods.
- `logRelF0-H1-A3_sma3nz_amean` → **Harmonic-Formant Diff (H1-A3)**: The arithmetic mean of the logarithmic energy ratio between the first fundamental frequency harmonic (H1) and the highest harmonic within the third formant range (A3).
- `F1frequency_sma3nz_amean` → **F1 Frequency (Avg)**: The arithmetic mean of

the first formant.